

K9-AIF Developer Guide

Architecture-First Development for Governed Agentic AI Systems

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1. Introduction

1.1 What K9-AIF Is

K9-AIF (K9 Agentic Intelligence Framework) is an architecture-first Python framework for building governed, observable, multi-agent AI systems at enterprise scale. It provides a layered set of abstract contracts (Architecture Building Blocks) and ready-to-use implementations (Solution Building Blocks) that teams can extend to build domain-specific AI solutions without starting from scratch and without sacrificing governance or observability.

The framework does not attempt to be a general-purpose AI toolkit. It is deliberately opinionated about structure: agents are small and focused, orchestrators coordinate squads of agents, routers decide which orchestrator handles an event, and governance runs at every layer boundary.

1.2 Why Architecture-First Agentic AI Matters

Most agentic AI systems begin as demos: a single agent connected to an LLM, returning answers. This works until the system needs to run in production, handle failures, respect data governance policies, produce audit trails, or be extended by a team. At that point, the absence of architectural structure becomes the primary source of risk.

K9-AIF applies a principle borrowed from enterprise architecture: define stable abstract contracts first, implement domain behavior second. This makes the system predictable, governable, and extensible without requiring rewrites.

1.3 Goals of the Framework

- Provide reusable ABBs that teams can extend without modifying the framework core
- Make governance, telemetry, and observability structural requirements, not afterthoughts
- Support iterative reasoning patterns (validation loops, actor-critic refinement) as first-class primitives
- Decouple model selection from agent logic through a routing layer
- Enable configuration-driven assembly of agents, squads, and orchestrators
- Support multiple inference backends, persistence stores, monitoring systems, and messaging backends through the factory pattern

1.4 ABB/SBB Philosophy

K9-AIF uses TOGAF-inspired terminology to distinguish between two kinds of components:

Architecture Building Blocks (ABBs) define what a component does. They are abstract classes that specify contracts — method signatures, lifecycle hooks, governance wiring — without implementing domain behavior. ABBs live in `k9_aif_abb/` and change infrequently.

Solution Building Blocks (SBBs) define how a specific solution implements those contracts. They extend ABBs with domain-specific logic, prompts, business rules, and tool integrations. SBBs live in `examples/` or `k9_projects/` and change frequently as requirements evolve.

This separation means the framework core stays stable while domain behavior evolves independently.

1.5 Router → Orchestrator → Squads → Agents Hierarchy

```
graph TD
    E[Inbound Event] --> R[Router]
    R --> O1[Orchestrator A]
    R --> O2[Orchestrator B]
    O1 --> S1[Squad 1]
    O1 --> S2[Squad 2]
    S1 --> A1[Agent 1]
    S1 --> A2[Agent 2]
    S1 --> A3[Agent 3]
    S2 --> A4[Agent 4]
    S2 --> A5[Agent 5]
```

Each layer has a single responsibility:

Layer	Responsibility	Key Contract
Router	Classify intent; select orchestrator	<code>route(payload)</code>
Orchestrator	Coordinate a domain workflow via squads	<code>execute_flow(payload)</code>
Squad	Execute agents in a defined sequence	<code>execute(payload)</code>
Agent	Perform one unit of work	<code>execute(payload)</code>

Each layer knows only what is below it. An agent does not know which squad it belongs to. A squad does not know which orchestrator called it. This decoupling is enforced by convention and by the YAML schema — agent YAML has no squad reference; squad YAML has no orchestrator reference.

1.6 Enterprise AI Systems vs Isolated Agents

Isolated agents solve demos. Enterprise AI systems solve production problems. The difference lies in governance (who authorized this action?), observability (what happened and why?), failure handling (what do we do when the LLM is unavailable?), and extensibility (how do we add a new domain without rewriting existing code?).

K9-AIF is designed for enterprise AI systems. Every architectural decision in the framework reflects these concerns.

2. Repository Orientation

2.1 Top-Level Repository Structure

```
k9-aif-framework/
├── k9_aif_abb/           # The framework package – ABBs and OOB implementations
├── examples/            # Reference SBB implementations (EOC is canonical)
├── k9_projects/         # Generated SBB stubs (from k9_generator.sh)
├── docs/                # Documentation
├── tests/               # Integration and smoke tests
├── k9_generator.sh      # Scaffold generator for new solutions
├── CLAUDE.md            # Claude Code integration guide
├── SKILLS.md            # Step-by-step development recipes
└── requirements.txt     # Python dependencies
```

2.2 k9_aif_abb Package Structure

```
k9_aif_abb/
├── k9_core/             # Foundation ABBs (abstract contracts)
│   ├── agent/           # BaseAgent, BaseMCPAgent
│   ├── router/          # BaseRouter
│   ├── orchestration/   # BaseOrchestrator, Handler (CoR)
│   ├── inference/       # BaseLLM, OllamaLLM, MockLLM
│   ├── governance/      # BaseGovernance, pipeline, require_governance()
│   ├── security/        # BaseSecurity, MockAuth
│   ├── persistence/     # BasePersistence
│   ├── messaging/       # BaseMessageAgent, K9EventBus
│   ├── monitoring/      # BaseMonitor, LoggerMonitor
│   ├── storage/         # BaseStorage
│   ├── retrieval/       # BaseRetriever, BaseDocParser, RetrieverRegistry
│   ├── integration/     # BaseConnector, MCPClientConnector, MCPHttpConnector
│   ├── streaming/       # BaseStreamProvider, RedpandaStreamProvider
│   ├── logging/         # BaseLogger
│   ├── formatter/       # BaseFormatter
│   ├── presentation/    # BaseUI
│   ├── iot/             # BaseIoTAgent
│   └── base_component.py # BaseComponent (root foundation)
├── k9_agents/           # Agent ABBs and OOB implementations
│   ├── validation/      # BaseValidationLoopAgent, K9ValidationLoopAgent
│   ├── critic_actor/    # BaseCriticActorAgent, K9CriticActorAgent
│   ├── chat/            # ChatAgentABB
│   ├── router/          # RouterAgent (intent classification)
│   ├── enrichment/     # EnrichmentAgent
│   ├── governance/      # GovernanceAgent
│   ├── security/        # AuthAgent, EncryptionAgent, SecretManagerAgent
│   ├── storage/         # FileStorageAgent, ObjectStorageAgent
│   └── messaging/       # KafkaAgent, SQSAgent, TopicMessageAgent
```

— integration/	# MCPClientAgent, WebSearchAgent
— retrieval/	# DoclingParser
— async_agent/	# AsyncAgent
— registry/	# AgentRegistry
— k9_squad/	# Squad orchestration
— base_squad.py	# BaseSquad
— squad_context.py	# SquadContext
— squad_loader.py	# SquadLoader
— squad_monitor.py	# Squad monitoring
— k9_orchestrators/	# OOB orchestrator implementations
— framework_orchestrator.py	
— governance_orchestrator.py	
— diagnostic_orchestrator.py	
— liveagent_orchestrator.py	
— orchestrator_loader.py	
— k9_inference/	# Model routing and inference
— models/	# InferenceRequest, InferenceResponse, RouteDecision
— routers/	# BaseModelRouter, K9ModelRouter, DefaultModelRouter
— catalog/	# ModelCatalog
— k9_factories/	# Component factories (LLM, Router, Monitor, etc.)
— k9_storage/	# Storage implementations (SQLite, PostgreSQL, File)
— k9_persistence/	# Persistence implementations
— k9_monitoring/	# Monitoring implementations (Prometheus, OTEL, etc.)
— k9_security/	# Zero-trust security layer
— k9_governance/	# Governance implementations (ProfanityGovernance)
— k9_data/	# Vector database and retrieval
— k9_mcp/	# MCP server implementations
— k9_adapters/	# Framework adapters (CrewAI)
— k9_utils/	# Utilities (config_loader, llm_invoke, timer)
— config/	# Framework default configuration YAML
— policies/	# Default governance policy YAML
— tests/	# Framework test suite

2.3 Major Folders and Responsibilities

Folder	What Belongs Here
k9_core/	Abstract contracts only. No domain logic, no concrete implementations except minimal utility classes like OllamaLLM and MockLLM
k9_agents/	Agent patterns (ABBs) and OOB agent implementations ready for domain extension

Folder	What Belongs Here
k9_squad/	Squad assembly, execution flow, and YAML loading
k9_orchestrators/	Concrete orchestrator implementations for framework-level concerns
k9_inference/	Model catalog, routing contracts, and the OOB K9ModelRouter
k9_factories/	All component construction; application code never instantiates framework components directly
k9_storage/ and k9_persistence/	Storage backends; agents and orchestrators depend on BasePersistence / BaseStorage, never on a concrete class
k9_security/	Zero-trust execution context model; wired optionally into routers and orchestrators
k9_utils/	Cross-cutting utilities; llm_invoke is the canonical LLM call path
config/	Framework-level default YAML. Do not treat this as production config. Each solution defines its own config/config.yaml

2.4 How Developers Should Navigate the Codebase

Start with the ABBs: Understand the contracts before reading implementations. The abstract classes in k9_core/ are concise and well-documented. Reading them takes less than an hour and reveals the full structure.

Read CLAUDE.md: This file contains the authoritative architecture notes, decoupling rules, and infrastructure defaults. It is more accurate than any documentation that lags the code.

Read SKILLS.md: This file contains step-by-step recipes for the most common tasks. Use it before writing any new agent, squad, orchestrator, or router.

Use the EOC example: examples/K9X_Enterprise_Insurance_OperationsCenter/ is the canonical reference implementation. When in doubt, read how EOC does it.

3. Core Architecture

3.1 BaseComponent

k9_core/base_component.py — the root foundation for all framework components.

```
class BaseComponent:
    def __init__(
```

```

        self,
        monitor=None,
        message_bus=None,
        config: Optional[Dict[str, Any]] = None,
    ):
        self.monitor = monitor
        self.logger = logging.getLogger(self.__class__.__name__)
        self.config = config or {}
        self.message_bus = message_bus

    async def log(self, message: str, level: str = "INFO", **kwargs) -> None:
        """Log to logger, emit to monitor, publish to message bus."""

```

BaseComponent wires together the three observability channels that every framework component uses: Python logger, monitor (for metrics), and message bus (for events). Most components inherit from it indirectly through BaseAgent, BaseOrchestrator, or BaseRouter.

3.2 BaseAgent

k9_core/agent/base_agent.py — the execution contract for every agent.

```

class BaseAgent(ABC):
    layer: str = "Agent Base"

    def __init__(
        self,
        config: Optional[Dict[str, Any]] = None,
        monitor=None,
        message_bus=None,
        governance=None,
    ):
        self.config = config or {}
        self.monitor = monitor
        self.message_bus = message_bus
        self.governance = require_governance(governance, self.config.get("k9_env"))
        self.logger = logging.getLogger(self.__class__.__name__)

    @abstractmethod
    def execute(self, payload: Dict[str, Any]) -> Dict[str, Any]:
        raise NotImplementedError("Subclasses must implement execute()")

    def publish_event(self, event: Dict[str, Any]) -> None:
        """Publish event to message bus and monitor."""

    async def apply_pre_governance(self, payload, ctx=None) -> Dict[str, Any]:
        """Apply governance pre-process hook before payload is sent outward."""

    async def apply_post_governance(self, payload, ctx=None) -> Dict[str, Any]:

```

```

        """Apply governance post-process hook after payload is received."""

    def enforce_governance(self) -> None:
        """
        Assert real governance is configured.
        - development/test: logs WARNING, continues
        - production/staging: raises PermissionError if NoopGovernance active
        """

```

The layer class attribute identifies the component in logs and events. Every subclass should override it with a meaningful name.

3.3 BaseRouter

k9_core/router/base_router.py — the intent routing contract.

```

class BaseRouter(ABC):
    layer: str = "Router Base"

    def __init__(
        self,
        config=None, monitor=None, message_bus=None, governance=None,
        zero_trust_guard=None, policy_enforcer=None,
        enable_zero_trust: Optional[bool] = None,
    ):
        # governance, zero-trust, registry wired at construction

    def register_orchestrator(self, intent: str, orchestrator: Any) -> None:
        """Register an orchestrator for a given intent."""

    @abstractmethod
    def route(self, payload: Dict[str, Any]) -> Dict[str, Any]:
        """Route the payload to the correct orchestrator."""

    def normalize(self, payload: Dict[str, Any]) -> Dict[str, Any]:
        """Normalize inbound payload before routing (optional hook)."""

    def apply_zero_trust(self, payload, ctx=None) -> Dict[str, Any]:
        """Apply zero-trust execution layer (if enabled)."""

```

The router is the entry point for all inbound events. It performs intent classification, optional zero-trust evaluation, and dispatches to the registered orchestrator. Routers own the first Kafka publish in a standard K9-AIF pipeline.

3.4 BaseOrchestrator

k9_core/orchestration/base_orchestrator.py — the workflow coordination contract.


```

class BaseOrchestrator(ABC):
    layer: str = "Orchestrator Base"

    def __init__(
        self,
        config=None, monitor=None, message_bus=None, governance=None,
        zero_trust_guard=None, policy_enforcer=None,
        enable_zero_trust: Optional[bool] = None,
    ):
        # governance, zero-trust wired at construction

    @abstractmethod
    def execute_flow(self, payload: Dict[str, Any]) -> Dict[str, Any]:
        """Execute the orchestration flow."""

    def publish_status(self, status: str, context: Dict[str, Any]) -> None:
        """Publish lifecycle status event."""

    def apply_zero_trust(self, payload, ctx=None) -> Dict[str, Any]:
        """Apply zero-trust evaluation if enabled."""

```

The orchestrator is responsible for the entire domain workflow. It loads and executes squads, applies governance, publishes status events, and returns the final result. It does not know about the router that dispatched to it.

3.5 BaseSquad

k9_squad/base_squad.py — sequential agent execution.

```

class BaseSquad:
    def __init__(self, squad_id, agents, orchestrator=None, monitor=None):
        self.squad_id = squad_id
        self.agents = agents or []
        self.flow = []

    def execute(self, payload: dict) -> dict:
        """
        Execute agents in flow order.
        Each step receives the accumulated context from all prior steps.
        Conditional steps (when:) are skipped if condition is false.
        Returns {"status": "completed", "squad_id": ..., **step_results}
        """

    def run(self, payload) -> dict:
        """Alias for execute()."""

```

BaseSquad is not abstract — it is a concrete implementation. Domain solutions rarely need to subclass it. The flow behavior is fully driven by the YAML definition loaded by SquadLoader.

3.6 Configuration-Driven Loading

K9-AIF uses YAML-driven assembly for agents, squads, and orchestrators. This means the structure of the system — which agents form a squad, what prompts they use, which model they invoke — is expressed in configuration rather than hardcoded.

The primary loaders are:

Loader	Input	Output
AgentRegistry	Name → class mapping	Agent instances
SquadLoader	YAML file + squad ID	BaseSquad with wired agents
OrchestratorLoader	Config dict	BaseOrchestrator instance
LLMFactory	Config block	Cached LLM instances
ModelRouterFactory	Config block	Cached router instance

3.7 Runtime Execution Model

sequenceDiagram

```
participant Client
participant Router
participant Orchestrator
participant Squad
participant Agent1
participant Agent2
participant LLM

Client->>Router: event payload
Router->>Router: classify intent
Router->>Orchestrator: dispatch payload
Orchestrator->>Squad: execute(payload)
Squad->>Agent1: execute(payload)
Agent1->>LLM: llm_invoke(request)
LLM-->>Agent1: InferenceResponse
Agent1-->>Squad: result dict
Squad->>Agent2: execute(payload + prior results)
Agent2-->>Squad: result dict
Squad-->>Orchestrator: accumulated context
Orchestrator-->>Client: final result
```

Context accumulation is the key concept: each agent in a squad enriches the shared context progressively. Agent 2 receives both the original payload and Agent 1's output. By the final agent, the context contains the full enriched state of the workflow.

3.8 Layering Strategy

```
graph TB
    subgraph "ABB Layer (k9_aif_abb)"
        BC[BaseComponent]
```

```

    BA[BaseAgent]
    BR[BaseRouter]
    B0[BaseOrchestrator]
    BS[BaseSquad]
    BVL[BaseValidationLoopAgent]
    BCA[BaseCriticActorAgent]
end

subgraph "OOB Layer (k9_aif_abb - implementations)"
    K9VL[K9ValidationLoopAgent]
    K9CA[K9CriticActorAgent]
    K9MR[K9ModelRouter]
end

subgraph "SBB Layer (examples / k9_projects)"
    DA[Domain Agent]
    D0[Domain Orchestrator]
    DR[Domain Router]
end

BA --> BVL
BA --> BCA
BVL --> K9VL
BCA --> K9CA
BA --> DA
B0 --> D0
BR --> DR
K9VL -.->|extend| DA
K9CA -.->|extend| DA

```

ABBs define contracts. OOB implementations provide ready-to-use defaults. Domain SBBs extend either the ABB directly (for full control) or the OOB implementation (to inherit defaults and override what differs).

4. ABB vs SBB Development Model

4.1 What Belongs in ABB

ABBs define stable, reusable contracts that are domain-agnostic. They belong in `k9_aif_abb/` and should:

- Contain only abstract methods or minimal lifecycle implementations
- Express architectural intent, not business rules
- Remain stable across releases
- Contain no references to specific domains (claims, fraud, medical)
- Contain no hardcoded prompts, business logic, or connection strings

Good ABB: `BaseValidationLoopAgent` — defines a hypothesis-validate-reason loop

skeleton. No domain logic.

Bad ABB: An agent that knows about insurance claims, contains specific prompts, or references a particular database schema.

4.2 What Belongs in SBB

SBBs realize ABB contracts for a specific solution. They belong in `examples/<App>/` or `k9_projects/<App>/` and should:

- Extend exactly one ABB (or OOB implementation)
- Contain all domain-specific prompts, business rules, and tool integrations
- Override only the methods required for their domain
- Not modify the parent ABB

Good SBB: `FraudDetectionAgent(K9ValidationLoopAgent)` — overrides `run_validation()` to call a fraud rule engine, overrides `should_continue()` for fraud-specific confidence thresholds.

4.3 What Belongs in Examples

The `examples/` directory contains complete, runnable reference implementations. The canonical example is `examples/K9X_Enterprise_Insurance_OperationsCenter/` (EOC). Examples demonstrate:

- How to wire agents into squads
- How to wire squads into orchestrators
- How to configure Kafka, PostgreSQL, governance policies
- How iterative patterns are used in practice

Examples are not templates. They are reference implementations to learn from, not copy.

4.4 Stable Contracts vs Domain Realizations

Aspect	ABB (Stable Contract)	SBB (Domain Realization)
Location	<code>k9_aif_abb/</code>	<code>examples/</code> or <code>k9_projects/</code>
Change rate	Infrequent	Frequent
Abstraction	Abstract class	Concrete class
Domain knowledge	None	Full
Prompt content	None	Yes
Business rules	None	Yes
Tests	Framework stability tests	Domain behavior tests

4.5 Anti-Patterns to Avoid

Anti-pattern: Putting domain logic in ABBs

```
# Wrong – domain knowledge in ABB
class BaseClaimsAgent(BaseAgent):
    COVERAGE_LIMIT = 50000 # domain constant has no place here
```

Anti-pattern: Bypassing the agent contract

```
# Wrong – calling Ollama directly from agent code
import requests
resp = requests.post("http://localhost:11434/api/generate", json={...})
```

Anti-pattern: Agents referencing squads

```
# Wrong – agent should never know its squad context
class MyAgent(BaseAgent):
    def execute(self, payload):
        next_agent = self.squad.agents[1] # forbidden coupling
```

Anti-pattern: Squad referencing its orchestrator

```
# Wrong – squad YAML must not reference its orchestrator
squads:
  MySquad:
    orchestrator: MyOrchestrator # this field must not exist
```

4.6 Why Not Everything Belongs in the Framework Core

The framework core is stable because it is small and focused. Every addition to the ABB layer increases the surface area that all solutions must track. Domain logic, prompt engineering, and business rules evolve at a much higher rate than architectural contracts. Keeping them in SBBs means the framework stays stable while solutions evolve freely.

5. Agent Development Guide

5.1 Creating a New Agent

Every new agent follows this structure:

Step 1 — Create the agent YAML

```
# examples/<App>/agents/yaml/my_assessment_agent.yaml

name: MyAssessmentAgent
class: MyAssessmentAgent

description: >
  Evaluates incoming requests and produces a structured assessment
  with confidence score.

pattern: reasoning
```

```

model: reasoning          # must match a key in inference.model_catalog

role: >
    You are a senior assessment specialist with deep expertise in
    evaluating complex requests against established criteria.

goal: >
    Analyze the provided input thoroughly. Return a structured assessment
    with a clear decision (approved/review/rejected) and a confidence score.

instructions:
    - Consider all dimensions of the request before deciding
    - Include specific reasoning for your decision
    - Confidence must reflect genuine certainty, not optimism
    - Always return valid JSON matching the output_schema

output_schema:
    decision: string (approved | review | rejected)
    reasoning: string
    confidence: float (0.0-1.0)
    flags: list[string]

governance:
    pre_process: true
    post_process: true

```

Step 2 — Create the Python class

```
# examples/<App>/agents/src/my_assessment_agent.py
```

```

import json
from typing import Any, Dict, Optional

from k9_aif_abb.k9_core.agent.base_agent import BaseAgent
from k9_aif_abb.k9_inference.models.inference_request import InferenceRequest
from k9_aif_abb.k9_utils.llm_invoke import llm_invoke

class MyAssessmentAgent(BaseAgent):

    layer = "<App> MyAssessmentAgent SBB"

    def __init__(self, config: Optional[Dict[str, Any]] = None, monitor=None, **kwargs):
        super().__init__(config or {}, monitor=monitor, **kwargs)

    def execute(self, payload: Dict[str, Any]) -> Dict[str, Any]:
        prompt = (
            f"Role: {self.config.get('role', '')}\n"
            f"Goal: {self.config.get('goal', '')}\n\n"

```

```

        f"Instructions:\n"
        + "\n".join(f"- {i}" for i in self.config.get("instructions", []))
        + f"\n\nInput:\n{json.dumps(payload, indent=2)}\n\n"
        f"Respond with valid JSON matching this schema:\n"
        f"{json.dumps(self.config.get('output_schema', {}), indent=2)}"
    )

    req = InferenceRequest(
        prompt=prompt,
        task_type=self.config.get("model", "general"),
        metadata={"agent": self.layer},
    )

    try:
        resp = llm_invoke(self.config, req)
    except RuntimeError as exc:
        self.logger.error("[%s] LLM unavailable: %s", self.layer, exc)
        return {"agent": self.layer, "output": "[WARN] LLM unavailable", "confidence": 0}

    self.publish_event({"type": "AssessmentCompleted", "agent": self.layer})

    return {
        "agent": self.layer,
        "output": resp.output.strip(),
        "model_used": resp.model_alias,
    }

```

Step 3 — Register in the orchestrator's `_load_squad()`

```

from examples.my_app.agents.src.my_assessment_agent import MyAssessmentAgent

agent_registry.register(
    "MyAssessmentAgent",
    lambda: MyAssessmentAgent(config=agent_loader.merge_with_global("MyAssessmentAgent"))
)

```

Step 4 — Add to the squad YAML flow

```

squads:
  AssessmentSquad:
    description: "Triage and assessment workflow."
    agents:
      - IntakeAgent
      - MyAssessmentAgent
      - AuditAgent
    flow:
      - agent: IntakeAgent
        result_key: intake
      - agent: MyAssessmentAgent
        result_key: assessment

```

```
- agent: AuditAgent
  result_key: audit
```

5.2 Agent YAML Configuration

Agent YAML files express all behavioral configuration. The fields available:

Field	Purpose	Required
name	Unique agent name	Yes
class	Python class name (must match exactly)	Yes
description	Human-readable purpose	Yes
pattern	Reasoning pattern: reasoning, extraction, chat, guardrails	Yes
model	Model catalog alias for LLM selection	Yes
role	LLM system prompt — who the agent is	Yes
goal	LLM user prompt — what to achieve	Yes
instructions	List of specific behavioral instructions	Recommended
output_schema	Expected output structure	Recommended
governance.pre_process	Apply governance before LLM call	Optional
governance.post_process	Apply governance after LLM call	Optional
max_tokens	Override model max tokens	Optional
confidence_threshold	For iterative agents	Optional

5.3 AgentLoader Behavior

When an orchestrator calls `agent_loader.merge_with_global(agent_name, global_config)`, the loader:

1. Reads the agent's YAML file
2. Merges with the global config.yaml
3. Agent YAML wins on key collision

The resulting merged dict is passed as `agent.config`. This means agents can access both their behavioral configuration (`self.config.get("role")`) and infrastructure configuration (`self.config.get("inference")`).

5.4 execute(payload) Contract

The `execute()` method is the agent's only public interface to the framework. It must:

- Accept a Dict[str, Any] payload
- Return a Dict[str, Any] result
- Be synchronous
- Handle its own LLM errors gracefully
- Never raise exceptions to the squad level for recoverable errors

The payload passed to an agent in a squad contains the original event plus all results from prior agents in the flow. Agents access prior results via standard dict keys.

5.5 Event Publishing Expectations

Agents should publish events for significant outcomes:

```
self.publish_event({
    "type": "AssessmentCompleted",
    "agent": self.layer,
    "correlation_id": payload.get("correlation_id"),
    "decision": result.get("decision"),
    "confidence": result.get("confidence"),
})
```

In standard K9-AIF solutions, agents are wired without a message_bus. Their publish_event() calls reach the monitor and logger only — not Kafka. This is intentional: agents share data sequentially through the squad flow, not through messaging.

5.6 Governance Integration Expectations

Agents that handle sensitive payloads should apply governance hooks:

```
import asyncio

def execute(self, payload):
    # Enforce governance is configured
    try:
        self.enforce_governance()
    except PermissionError as exc:
        self.logger.error("[%s] %s", self.layer, exc)
        return {"agent": self.layer, "output": "[WARN] governance not configured"}

    # Apply pre-governance (sanitize input)
    payload = asyncio.get_event_loop().run_until_complete(
        self.apply_pre_governance(payload)
    )

    # ... LLM call ...

    # Apply post-governance (validate output)
    result = asyncio.get_event_loop().run_until_complete(
        self.apply_post_governance(result)
```

```
)

return result
```

5.7 Do / Do Not

Do	Do Not
Extend BaseAgent	Call OllamaLLM directly
Use llm_invoke() for all LLM calls	Import LLMFactory in agent code
Handle RuntimeError from llm_invoke	Let LLM errors propagate as exceptions
Set layer class attribute	Omit layer
Publish events for significant outcomes	Publish raw payload data in events
Use self.config.get("role") for prompts	Hardcode prompts in Python
Keep agents focused on one concern	Build multi-concern agents

5.8 Choosing Between Agent Patterns

```
graph TD
    Q1{Does the agent need to test<br/>something, observe the result,<br/>and decide wh
    Q2{Does the agent produce and<br/>iteratively refine an output<br/>through critique
    A1[Use BaseAgent<br/>one-shot execution]
    A2[Use BaseValidationLoopAgent<br/>or K9ValidationLoopAgent]
    A3[Use BaseCriticActorAgent<br/>or K9CriticActorAgent]

    Q1 -->|No| Q2
    Q1 -->|Yes| A2
    Q2 -->|No| A1
    Q2 -->|Yes| A3
```

Solution Architect Note: The generator scaffolds all agents as BaseAgent by default. The SA must explicitly decide at design time which agents need iterative behavior. Most agents — triage, routing, audit, guard, graph sync — are correctly one-shot. Reserve iterative patterns for agents that genuinely need to converge on confidence.

6. Orchestrator Development Guide

6.1 When to Create an Orchestrator

Create a new orchestrator when you have a distinct domain workflow that requires coordinating multiple squads or applying domain-specific governance. Do not create an orchestrator for every feature — one orchestrator per domain workflow is the right granularity.

6.2 Orchestrator Responsibilities

A K9-AIF orchestrator is responsible for:

1. Loading and executing the appropriate squad(s) for the workflow
2. Applying pre/post governance at the workflow boundary
3. Optionally applying zero-trust evaluation
4. Publishing status events for the lifecycle (started, completed, failed)
5. Returning the final result to the router or caller

An orchestrator is **not** responsible for:

- Routing decisions (that belongs to the Router)
- Individual agent execution (that belongs to the Squad)
- LLM invocation (that belongs to the Agent via `llm_invoke`)

6.3 Squad Coordination

```
class MyDomainOrchestrator(BaseOrchestrator):
```

```
    layer = "MyDomain Orchestrator SBB"  
    _SQUAD_ID = "MyDomainSquad"
```

```
    def __init__(self, config=None, monitor=None, message_bus=None, governance=None):  
        super().__init__(config or {}, monitor=monitor,  
                          message_bus=message_bus, governance=governance)  
        self.squad = self._load_squad()
```

```
    def execute_flow(self, payload: Dict[str, Any]) -> Dict[str, Any]:  
        self.publish_status("started", {"squad_id": self._SQUAD_ID,  
                                         "correlation_id": payload.get("correlation_id")  
        try:  
            result = self.squad.execute(payload)  
            self.publish_status("completed", {"squad_id": self._SQUAD_ID,  
                                              "status": result.get("status")})  
            return result  
        except Exception as exc:  
            self.logger.error("[%s] Flow failed: %s", self.layer, exc)  
            self.publish_status("failed", {"error": str(exc)})  
            return {"status": "error", "error": str(exc)}
```

```
    def _load_squad(self) -> BaseSquad:  
        from k9_aif_abb.k9_squad.squad_loader import SquadLoader  
        from k9_aif_abb.k9_agents.registry.agent_registry import AgentRegistry  
        # register agents and load squad  
        agent_registry = AgentRegistry()  
        # ... register agents ...  
        loader = SquadLoader(agent_registry)  
        return loader.load_one("config/squads.yaml", self._SQUAD_ID)
```

6.4 Routing Boundaries

The orchestrator's boundary is clear:

- **Receives from:** The Router via `execute_flow(payload)`
- **Delegates to:** One or more Squads via `squad.execute(payload)`
- **Returns to:** The Router (or the Kafka result topic)

The orchestrator does not need to know how the payload arrived. The payload should contain everything the workflow needs, stamped by the router before dispatch.

6.5 Failure Handling

Orchestrators should catch all exceptions from squad execution and return a structured error result rather than propagating the exception. This prevents one domain failure from crashing the entire routing process.

```
try:
    result = self.squad.execute(payload)
except Exception as exc:
    self.logger.error("[%s] Squad execution failed: %s", self.layer, exc)
    self.publish_status("failed", {"reason": str(exc)})
    return {
        "status": "error",
        "domain": "my_domain",
        "error": str(exc),
        "correlation_id": payload.get("correlation_id"),
    }
```

6.6 Enterprise Orchestration Principles

- **One orchestrator per domain workflow.** Do not create orchestrators for sub-tasks.
 - **Publish lifecycle events.** Use `publish_status()` at start, completion, and failure.
 - **Preserve correlation IDs.** Pass `correlation_id` through all events for distributed tracing.
 - **Keep orchestrators stateless.** Load squads at startup; do not maintain execution state between calls.
 - **Never bypass governance.** Apply `apply_pre_governance()` and `apply_post_governance()` for any regulated domain.
-

7. Router Development Guide

7.1 Intent Routing

The router is the framework entry point. In a Kafka-based deployment, it consumes from the primary event topic and publishes to domain-specific topics. In a direct-call deployment, it selects and invokes the correct orchestrator.

The router's decision logic should be deterministic where possible:

- **Deterministic routing:** event_type field maps directly to an orchestrator
- **Intent-driven routing:** LLM or classifier determines intent, then maps to orchestrator

7.2 Router Responsibilities

Responsibility	Notes
Intent classification	Deterministic or LLM-based
Orchestrator selection	Via register_orchestrator() registry
Zero-trust evaluation	Optional, via apply_zero_trust()
Governance pre-processing	Optional, via apply_pre_governance()
Event publishing	Publish routing decision for audit trail
Payload normalization	Stamp common fields (correlation_id, intent)

7.3 Implementing a Router

```
class MyRouter(BaseRouter):
```

```
    layer = "MyRouter SBB"
```

```
    def __init__(self, config=None, monitor=None, message_bus=None, governance=None):
        super().__init__(config or {}, monitor=monitor,
                         message_bus=message_bus, governance=governance)
        self._setup_orchestrators()
```

```
    def route(self, payload: Dict[str, Any]) -> Dict[str, Any]:
        event_type = payload.get("event_type", "unknown")
        payload = self.normalize(payload)
```

```
        orchestrator = self.registry.get(event_type)
```

```
        if not orchestrator:
            self.logger.warning("[%s] No orchestrator for event_type: %s", self.layer,
                               event_type)
            return {"status": "unrouted", "event_type": event_type}
```

```
        self.publish_event({
            "type": "EventRouted",
            "event_type": event_type,
            "correlation_id": payload.get("correlation_id"),
        })
```

```
        return orchestrator.execute_flow(payload)
```

```
    def normalize(self, payload: Dict[str, Any]) -> Dict[str, Any]:
        import uuid
```

```

        payload.setdefault("correlation_id", str(uuid.uuid4()))
        return payload

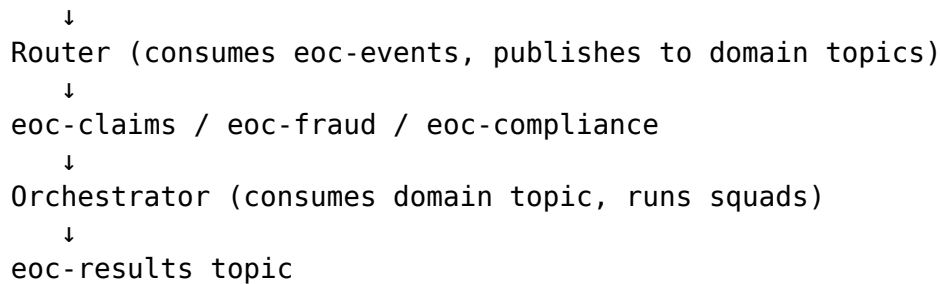
    def _setup_orchestrators(self):
        from examples.my_app.orchestrators.claims_orchestrator import ClaimsOrchestrator
        from examples.my_app.orchestrators.fraud_orchestrator import FraudOrchestrator
        self.register_orchestrator("claims_event", ClaimsOrchestrator(self.config))
        self.register_orchestrator("fraud_alert", FraudOrchestrator(self.config))

```

7.4 Kafka Event Topology

In a Kafka-based deployment:

Client → eoc-events topic



Kafka ownership rule: The Router is the only publisher of domain topics. The Orchestrator is the only consumer of domain topics and publisher of results. Agents never publish to Kafka in a standard K9-AIF solution.

7.5 Router-to-Orchestrator Handoff

The handoff is a method call (synchronous) or Kafka topic publish (async). In both cases, the payload contains:

- event_type — what happened
- correlation_id — for distributed tracing
- intent — if LLM classification was applied
- All original domain data

The orchestrator receives this payload and begins `execute_flow()` without knowledge of how it was dispatched.

8. Validation Loop Pattern

8.1 Overview

The Validation Loop is an iterative reasoning pattern for agents that must test a hypothesis, observe the result, and decide whether to continue before producing a final answer.

```

graph TD
    S[execute payload] --> GH[generate_hypothesis]
    GH --> RV[run_validation]
    RV --> EO[evaluate_observation]
    EO --> SC{should_continue?}
    SC -->|CONTINUE| GH
    SC -->|FINALIZE| F[finalize]
    SC -->|ESCALATE| E[escalate]
    SC -->|FAIL| FA[fail]
    F --> R[return dict]
    E --> R
    FA --> R

```

8.2 BaseValidationLoopAgent

k9_agents/validation/base_validation_loop_agent.py

The ABB that provides the loop skeleton. Five abstract methods define the domain behavior:

```

class BaseValidationLoopAgent(BaseAgent):
    layer: str = "BaseValidationLoopAgent"

    # Loop skeleton is implemented in execute() – do not override it

    @abstractmethod
    def generate_hypothesis(self, loop_ctx: ValidationLoopContext) -> Any:
        """Form the next thing to test. Has access to loop_ctx.steps (prior iterations)"""

    @abstractmethod
    def run_validation(self, hypothesis: Any, loop_ctx: ValidationLoopContext) -> Any:
        """Invoke tool/function/rule engine/LLM to test the hypothesis."""

    @abstractmethod
    def evaluate_observation(self, tool_result: Any, loop_ctx: ValidationLoopContext) -> Any:
        """Interpret raw tool result. Return dict with 'confidence' key (float 0.0-1.0)"""

    @abstractmethod
    def should_continue(self, observation: Any, loop_ctx: ValidationLoopContext) -> ValidationDisposition:
        """Return CONTINUE | FINALIZE | ESCALATE | FAIL."""

    @abstractmethod
    def finalize(self, loop_ctx: ValidationLoopContext) -> ValidationLoopResult:
        """Produce the validated output."""

    # Optional overrides – defaults provided
    def escalate(self, loop_ctx: ValidationLoopContext) -> ValidationLoopResult:
        """Default: return ESCALATE disposition. Override for domain HIL logic."""

```

```
def fail(self, loop_ctx: ValidationLoopContext) -> ValidationLoopResult:
    """Default: return FAIL disposition. Override for domain failure output."""
```

Config keys for tuning the loop:

```
max_iterations: 5                # hard cap on iterations (default: 5)
confidence_threshold: 0.8        # available to should_continue() (default: 0.8)
finalize_on_max_iterations: true  # true → finalize; false → escalate on timeout
escalate_on_tool_error: false    # true → ESCALATE; false → FAIL on run_validation()
```

8.3 K9ValidationLoopAgent

k9_agents/validation/k9_validation_loop_agent.py

The OOB implementation where the LLM is the validation tool. Use this when LLM reasoning is sufficient for validation.

```
class K9ValidationLoopAgent(BaseValidationLoopAgent):
    """OOB: LLM is the validation tool."""

    def generate_hypothesis(self, loop_ctx: ValidationLoopContext) -> str:
        """Build prompt from payload + prior iterations."""

    def run_validation(self, hypothesis: str, loop_ctx: ValidationLoopContext) -> str:
        """Call llm_invoke() – LLM validates the hypothesis."""

    def evaluate_observation(self, tool_result: str, loop_ctx: ValidationLoopContext) -> str:
        """Parse JSON from LLM response."""

    def should_continue(self, observation: Dict, loop_ctx: ValidationLoopContext) -> ValidationDisposition:
        """Compare confidence vs threshold; check needs_more signal."""

    def finalize(self, loop_ctx: ValidationLoopContext) -> ValidationLoopResult:
        """Package last observation + step history."""
```

8.4 ValidationDisposition

```
class ValidationDisposition(str, Enum):
    CONTINUE = "continue"    # Loop continues – insufficient confidence
    FINALIZE = "finalize"    # Confidence sufficient – produce output
    ESCALATE = "escalate"    # Uncertain – route to human-in-the-loop
    FAIL     = "fail"        # Definitive negative result
```

8.5 ValidationLoopContext and Result

```
@dataclass
class ValidationLoopContext:
    payload: Dict[str, Any]
    steps: List[ValidationLoopStep] # history of all prior iterations
    iteration: int                  # current iteration count
```



```

        metadata: Dict[str, Any]

@dataclass
class ValidationLoopStep:
    iteration: int
    hypothesis: Any
    tool_result: Any
    observation: Any
    disposition: ValidationDisposition
    confidence: float

@dataclass
class ValidationLoopResult:
    disposition: ValidationDisposition
    output: Dict[str, Any]
    steps: List[ValidationLoopStep]
    iterations: int
    final_confidence: float
    evidence: List[str]

```

8.6 Custom Domain Validation Agent Example

Extending K9ValidationLoopAgent for fraud detection — overriding only what differs:

```

from k9_aif_abb.k9_agents.validation.k9_validation_loop_agent import K9ValidationLoopAgent
from k9_aif_abb.k9_agents.validation.models.validation_loop import (
    ValidationDisposition, ValidationLoopContext, ValidationLoopResult
)

class FraudDetectionAgent(K9ValidationLoopAgent):

    layer = "EOC FraudDetectionAgent SBB"

    def run_validation(self, hypothesis: str, loop_ctx: ValidationLoopContext):
        # Replace LLM-only validation with domain rule engine
        from examples.eoc.rules.fraud_rules import FraudRuleEngine
        return FraudRuleEngine().evaluate(loop_ctx.payload)

    def should_continue(self, observation: dict, loop_ctx: ValidationLoopContext):
        confidence = observation.get("confidence", 0.0)
        if confidence >= 0.9:
            return ValidationDisposition.FINALIZE
        if confidence < 0.2 and loop_ctx.iteration >= 2:
            return ValidationDisposition.FAIL
        if loop_ctx.iteration >= 3 and confidence < 0.5:
            return ValidationDisposition.ESCALATE
        return ValidationDisposition.CONTINUE

```

```

def finalize(self, loop_ctx: ValidationLoopContext) -> ValidationLoopResult:
    last = loop_ctx.steps[-1]
    return ValidationLoopResult(
        disposition=ValidationDisposition.FINALIZE,
        output={
            "fraud_determination": last.observation.get("decision"),
            "confidence": last.confidence,
            "risk_score": last.observation.get("risk_score"),
        },
        steps=loop_ctx.steps,
        iterations=loop_ctx.iteration,
        final_confidence=last.confidence,
        evidence=[str(s.observation) for s in loop_ctx.steps],
    )

```

8.7 Extending from BaseValidationLoopAgent Directly

When the LLM is not the validation tool — for example, using a rule engine, database query, or external API:

```

class ComplianceGapAgent(BaseValidationLoopAgent):

```

```

    layer = "ComplianceGapAgent SBB"

```

```

def generate_hypothesis(self, loop_ctx: ValidationLoopContext):
    prior_gaps = [s.observation.get("unchecked_clauses", []) for s in loop_ctx.steps]
    remaining = [c for c in loop_ctx.payload.get("clauses", [])
                 if c not in [g for gaps in prior_gaps for g in gaps]]
    return {"clauses_to_check": remaining[:3]} # check 3 clauses per iteration

```

```

def run_validation(self, hypothesis, loop_ctx: ValidationLoopContext):
    return compliance_db.check_clauses(hypothesis["clauses_to_check"])

```

```

def evaluate_observation(self, tool_result, loop_ctx: ValidationLoopContext):
    gaps = [r for r in tool_result if not r["compliant"]]
    confidence = 1.0 - (len(gaps) / max(len(tool_result), 1))
    return {"gaps": gaps, "checked": [r["clause"] for r in tool_result], "confidence": confidence}

```

```

def should_continue(self, observation, loop_ctx: ValidationLoopContext):
    all_clauses = loop_ctx.payload.get("clauses", [])
    checked = sum(len(s.observation.get("checked", [])) for s in loop_ctx.steps)
    checked += len(observation.get("checked", []))
    if checked >= len(all_clauses):
        return ValidationDisposition.FINALIZE
    return ValidationDisposition.CONTINUE

```

```

def finalize(self, loop_ctx: ValidationLoopContext) -> ValidationLoopResult:

```

```

all_gaps = [g for s in loop_ctx.steps for g in s.observation.get("gaps", [])]
return ValidationLoopResult(
    disposition=ValidationDisposition.FINALIZE,
    output={"compliance_gaps": all_gaps, "gap_count": len(all_gaps)},
    steps=loop_ctx.steps,
    iterations=loop_ctx.iteration,
    final_confidence=1.0 if not all_gaps else 0.6,
    evidence=[str(g) for g in all_gaps],
)

```

8.8 Telemetry Events

BaseValidationLoopAgent emits these events via `publish_event()` at each step:

Event Type	When Emitted
loop_started	Once, at <code>execute()</code> entry
hypothesis_generated	Each iteration
validation_tool_invoked	After successful <code>run_validation()</code>
observation_evaluated	After <code>evaluate_observation()</code>
loop_continued	When disposition is CONTINUE
loop_finalized	When disposition is FINALIZE
loop_escalated	When disposition is ESCALATE
loop_failed	When disposition is FAIL or tool error

8.9 Use Cases

Domain	Hypothesis	Validation Tool	Convergence Signal
Fraud detection	Fraud signals to correlate	Rule engine + velocity checks	Risk score threshold
Claims processing	Coverage clauses to check	Policy database	All clauses reviewed
Compliance	Regulatory gaps to assess	Compliance database	No unchecked requirements
Document extraction	Schema fields to extract	OCR + schema validator	All required fields populated
Security	Vulnerabilities to confirm	Static analysis + exploit check	Confidence in CVE presence

8.10 When Iterative Reasoning Is Architecturally Appropriate

Use the validation loop only when:

1. The problem requires testing a hypothesis against evidence that changes with each iteration

2. A single-pass answer is insufficient because confidence accumulates over multiple checks
3. The agent must decide at each step whether more evidence is needed
4. Escalation to human review is a legitimate outcome

Do not use the validation loop for:

- Simple classification or triage tasks
- Agents that call the LLM once and return
- Tasks where the answer does not depend on accumulated evidence
- Routing, audit, or guard agents

9. Critic-Actor Pattern

9.1 Overview

The Critic-Actor pattern produces iteratively refined output. The Actor generates a draft; the Critic evaluates it and provides structured feedback; the Actor refines using that feedback. The loop continues until the Critic accepts the output or a terminal condition is reached.

```
graph TD
    S[execute payload] --> G[generate Round 1]
    G --> C[critique]
    C --> SA{should_accept?}
    SA -->|REJECTED| R[refine Round N]
    R --> C
    SA -->|ACCEPTED| F[finalize]
    SA -->|ESCALATE| E[escalate]
    SA -->|FAIL| FA[fail]
    F --> RES[return dict]
    E --> RES
    FA --> RES
```

9.2 BaseCriticActorAgent

k9_agents/critic_actor/base_critic_actor_agent.py

```
class BaseCriticActorAgent(BaseAgent):
    layer: str = "BaseCriticActorAgent"

    @abstractmethod
    def generate(self, ctx: CriticActorContext) -> Any:
        """Actor: produce initial draft. Called on round 1 only."""

    @abstractmethod
    def critique(self, draft: Any, ctx: CriticActorContext) -> Dict[str, Any]:
        """Critic: evaluate draft. Return {accepted: bool, score: float, issues: list}.
```

```

@abstractmethod
def refine(self, draft: Any, feedback: Dict[str, Any], ctx: CriticActorContext) ->
    """Actor: improve draft using Critic's feedback. Called on rounds 2+."""

@abstractmethod
def should_accept(self, feedback: Dict[str, Any], ctx: CriticActorContext) -> Criti
    """Return ACCEPTED | REJECTED | ESCALATE | FAIL."""

@abstractmethod
def finalize(self, ctx: CriticActorContext) -> CriticActorResult:
    """Produce final accepted output."""

def escalate(self, ctx: CriticActorContext) -> CriticActorResult: ...
def fail(self, ctx: CriticActorContext) -> CriticActorResult: ...

```

Config keys:

```

max_rounds: 3                # hard cap on refinement rounds (default: 3)
acceptance_threshold: 0.8    # score threshold for acceptance (default: 0.8)
finalize_on_max_rounds: true # true → finalize; false → escalate on timeout
escalate_on_critic_error: false # true → ESCALATE; false → FAIL on critique() error

```

9.3 K9CriticActorAgent

k9_agents/critic_actor/k9_critic_actor_agent.py

OOB implementation where the LLM plays both Actor and Critic roles.

```

class K9CriticActorAgent(BaseCriticActorAgent):
    """OOB: LLM plays both Actor and Critic."""

    def generate(self, ctx: CriticActorContext) -> str:
        """Actor LLM call: role/goal + payload → initial draft."""

    def critique(self, draft: str, ctx: CriticActorContext) -> Dict[str, Any]:
        """Critic LLM call: critic_role/goal + draft → JSON feedback."""

    def refine(self, draft: str, feedback: Dict[str, Any], ctx: CriticActorContext) ->
        """Actor LLM call: role/goal + draft + issues → improved draft."""

    def should_accept(self, feedback: Dict[str, Any], ctx: CriticActorContext) -> Criti
        """Check feedback['accepted'] and score >= acceptance_threshold."""

    def finalize(self, ctx: CriticActorContext) -> CriticActorResult:
        """Package accepted draft + round history."""

```

9.4 CriticActorDisposition

```
class CriticActorDisposition(str, Enum):
    ACCEPTED = "accepted"    # Critic satisfied – finalize
    REJECTED = "rejected"    # Critic found issues – refine
    ESCALATE = "escalate"    # Cannot converge – route to HIL
    FAIL     = "fail"        # Definitively unacceptable
```

9.5 CriticActorContext and Result

```
@dataclass
class CriticActorContext:
    payload: Dict[str, Any]
    steps: List[CriticActorStep] # history of all rounds
    round: int                  # current round number
    metadata: Dict[str, Any]

@dataclass
class CriticActorStep:
    round: int
    draft: Any
    feedback: Dict[str, Any]    # {accepted, score, issues, summary}
    disposition: CriticActorDisposition
    score: float

@dataclass
class CriticActorResult:
    disposition: CriticActorDisposition
    output: Dict[str, Any]
    steps: List[CriticActorStep]
    rounds: int
    final_score: float
    critique_log: List[str]
```

9.6 Difference Between Validation Loop and Critic-Actor

Aspect	Validation Loop	Critic-Actor
Primary concern	Convergence on evidence-backed truth	Quality refinement of output
Roles	One agent: hypothesis + observe	Two roles: Actor generates, Critic evaluates
Iteration trigger	Insufficient confidence in observation	Critic rejected draft
Termination condition	Confidence threshold	Critic accepts OR rounds exhausted
Typical outcome	Decision (approved/rejected/escalated)	Refined artifact (document/schema/report)

Aspect	Validation Loop	Critic-Actor
Best use cases	Fraud, compliance, evidence review	Contract drafting, schema refinement, report writing

9.7 Plugging in a Real Critic

The most powerful use of BaseCriticActorAgent is substituting a real external critic for the LLM critic:

```
class ContractDraftingAgent(K9CriticActorAgent):

    layer = "ContractDraftingAgent SBB"

    def critique(self, draft: str, ctx: CriticActorContext) -> Dict[str, Any]:
        # Use real compliance checker instead of LLM critic
        issues = legal_compliance_checker.validate(draft)
        schema_ok = contract_schema_validator.check(draft)
        score = 1.0 if not issues and schema_ok else max(0.3, 1.0 - 0.2 * len(issues))
        return {
            "accepted": not issues and schema_ok,
            "score": score,
            "issues": issues,
            "schema_valid": schema_ok,
            "summary": f"{len(issues)} compliance issues found.",
        }
```

9.8 Telemetry Events

Event Type	When Emitted
loop_started	Once, at execute() entry
draft_generated	After generate() on round 1
draft_refined	After refine() on rounds 2+
critique_produced	After successful critique()
loop_accepted	When disposition is ACCEPTED
loop_rejected	When disposition is REJECTED
loop_escalated	When disposition is ESCALATE
loop_failed	When disposition is FAIL or critic error

9.9 Use Cases

Domain	Actor Task	Critic	Refinement Target
Contract drafting	Draft legal clause	Compliance checker	Zero legal issues

Domain	Actor Task	Critic	Refinement Target
Schema refinement	Generate JSON schema	Schema validator	Valid, complete schema
Report improvement	Draft executive report	Quality rubric	Meets quality standard
Policy review	Draft policy statement	Regulatory checker	Regulatory compliance
Code generation	Write function	Test runner	All tests pass

10. Planning Loop Pattern

10.1 When to Use K9PlanningLoopAgent

Use K9PlanningLoopAgent when the agent must **plan its own steps and revise the plan as it goes**. Unlike K9ValidationLoopAgent (which converges on a confidence score), the planning loop maintains a dynamic plan (`remaining_steps`) and a scratchpad (notes) that evolve across iterations.

Decision rule:

One-pass	Validation Loop	Planning Loop
Classify, route, audit	Fraud correlation, document confidence	Investigation, multi-stage research, architecture planning
BaseAgent	K9ValidationLoopAgent	K9PlanningLoopAgent

10.2 How It Works

K9PlanningLoopAgent extends BaseValidationLoopAgent. Each iteration, the LLM is shown its current plan and scratchpad. It returns: - An updated `remaining_steps` list - Updated notes (scratchpad) - confidence and reasoning

The loop finalizes when: - The LLM returns an empty `remaining_steps` (plan complete), OR - confidence reaches `confidence_threshold`

If the LLM's response cannot be parsed, behavior falls back to confidence-driven continuation exactly like K9ValidationLoopAgent.

10.3 Implementation

```
from k9_aif_abb.k9_agents.planning import K9PlanningLoopAgent
```

```
class ArchitecturePlannerAgent(K9PlanningLoopAgent):
```

```
    layer = "ArchitecturePlannerAgent SBB"
```



```

def should_continue(self, observation, loop_ctx):
    if observation["confidence"] < 0.2:
        return ValidationDisposition.FAIL
    return super().should_continue(observation, loop_ctx)

```

10.4 Configuration

```

name: ArchitecturePlannerAgent
class: ArchitecturePlannerAgent
pattern: reasoning
model: reasoning
max_iterations: 8
confidence_threshold: 0.85
finalize_on_max_iterations: true

```

role: >

You are an enterprise architect that generates architecture plans.

goal: >

Generate a complete, actionable architecture plan with clear steps.

10.5 Output

K9PlanningLoopAgent._to_dict() includes two additional keys beyond the standard BaseValidationLoopAgent output:

Key	Type	Meaning
remaining_steps	list[str]	Final plan state at finalize time (empty if complete)
notes	dict	Final scratchpad state

10.6 Inheritance Hierarchy

```

BaseAgent
├── BaseValidationLoopAgent    (loop skeleton – ABB)
│   ├── K9ValidationLoopAgent (confidence convergence – 00B)
│   └── K9PlanningLoopAgent   (dynamic plan + scratchpad – 00B)

```

11. Prompt Evaluation Pattern

11.1 Overview

The Prompt Evaluation Pattern provides a development-time pipeline for grading authored prompts before they enter a workflow. It answers the question: how well does this prompt perform across a range of inputs, and what grade does it earn?

Scope: This is a design-time and measurement-time tool. It grades authored prompts — system prompts, agent instructions, guided-flow templates — not user-provided inputs at runtime. Runtime quality enforcement is K9ValidationLoopAgent’s responsibility.

graph LR

```
A[Author Prompt] --> B[Define Test Cases]
B --> C[run_suite]
C --> D{Grade >= C?}
D -->|Yes| E[Promote to Config]
D -->|No| F[Review Rationale]
F --> G[Revise Prompt]
G --> H[compare variants]
H --> I[Adopt Winner]
I --> C
```

11.2 BasePromptEvaluator — ABB Contract

k9_core/evaluation/base_prompt_evaluator.py

Three abstract methods define the contract:

```
class BasePromptEvaluator(BaseComponent, ABC):
    layer: str = "BasePromptEvaluator"

    @abstractmethod
    def evaluate(
        self,
        prompt: str,
        input_data: Dict[str, Any],
        actual_output: str,
        expected: str,
        test_case_description: str = "",
    ) -> EvaluationResult:
        """Score a single prompt execution against an expectation."""

    @abstractmethod
    def compare(
        self,
        prompt_a: str,
        prompt_b: str,
        test_cases: List[PromptTestCase],
    ) -> ComparisonResult:
        """A/B test two prompt variants across a list of test cases."""

    @abstractmethod
    def run_suite(
        self,
        prompt: str,
        test_cases: List[PromptTestCase],
```

```
) -> SuiteResult:
    """Batch evaluation across a list of test cases."""
```

11.3 Evaluation Data Models

```
@dataclass
```

```
class EvaluationResult:
    score: float          # 0-100 composite score
    grade: str            # A | B | C | D | F
    verdict: str          # PASS | FAIL
    dimensions: List[DimensionScore]
    rationale: str        # judge's overall summary
    actual_output: str
    prompt: str
    test_case_description: str
```

```
@dataclass
```

```
class DimensionScore:
    name: str            # correctness | completeness | format_compliance | clarity | relevancy
    score: float         # 0-100
    rationale: str
```

```
@dataclass
```

```
class ComparisonResult:
    winner: str          # prompt_a | prompt_b | tie
    score_a: float
    score_b: float
    grade_a: str
    grade_b: str
    rationale: str
    results_a: List[EvaluationResult]
    results_b: List[EvaluationResult]
```

```
@dataclass
```

```
class SuiteResult:
    total: int
    passed: int
    failed: int
    average_score: float
    overall_grade: str
    pass_rate: float
    results: List[EvaluationResult]
```

```
@dataclass
```

```
class PromptTestCase:
    input_data: Dict[str, Any]
    expected: str
    description: str = ""
```

11.4 K9PromptEvaluator — OOB SBB

k9_agents/evaluation/k9_prompt_evaluator.py

The OOB implementation uses LLM-as-judge via `llm_invoke()`. No external evaluation service; no additional dependencies beyond the inference layer already present.

The judge scores five weighted dimensions:

Dimension	Weight	Question
Correctness	35%	Does the output correctly answer the task?
Completeness	25%	Does it cover all required aspects?
Format compliance	15%	Does it follow the requested format / structure?
Clarity	15%	Is the output clear, coherent, and readable?
Relevance	10%	Is the output focused and on-topic?

Grade scale: A (90+), B (80–89), C (70–79), D (60–69), F (<60). Default PASS threshold: 70.

Key design: Each LLM call sets `metadata["operation"]` to separate concerns in telemetry and routing:

- "operation": "invoke" — the prompt execution call (generates the output to evaluate)
- "operation": "evaluate" — the judge call (scores the output)

This keeps evaluation traffic identifiably separate from production inference traffic.

11.5 EvaluationFactory

EvaluationFactory follows the standard K9-AIF factory pattern:

```
from k9_aif_abb.k9_factories.evaluation_factory import EvaluationFactory

evaluator = EvaluationFactory.create(config)
# Returns K9PromptEvaluator by default (provider: k9)
# Returns custom SBB when provider: my_evaluator is set

evaluation:
    provider: k9          # default OOB – K9PromptEvaluator
    pass_threshold: 70    # PASS when score >= this
    judge_model: reasoning
```

11.6 Developer Workflow

The workflow from authored prompt to promoted config:

1. **Author** the prompt template (system prompt, agent instructions, guided-flow step)
2. **Define test cases** — representative inputs and expected output behaviour
3. **Run `run_suite()`** — batch score across all test cases
4. **Review results** — inspect failing cases; identify weak dimensions
5. **Compare variants** — use `compare()` to A/B test the revised prompt against the original
6. **Adopt the winner** and re-run the suite to confirm the improvement holds

When the underlying model changes (upgrade or provider swap), re-run the suite. The scores either hold or they do not — that delta is the signal.

11.7 Extending the Evaluator

K9PromptEvaluator is one implementation. Solution Architects extend BasePromptEvaluator for domain-specific evaluation needs:

```
from k9_aif_abb.k9_core.evaluation.base_prompt_evaluator import BasePromptEvaluator

class ClinicalPrecisionEvaluator(BasePromptEvaluator):
    """Domain-calibrated evaluator for clinical AI prompts."""
    layer = "ClinicalPrecisionEvaluator SBB"

    def evaluate(self, prompt, input_data, actual_output, expected, description=""):
        # Replace generic dimensions with clinical rubrics:
        # clinical_accuracy, drug_safety, terminology_correctness,
        # evidence_citation, format
        ...

    def compare(self, prompt_a, prompt_b, test_cases): ...
    def run_suite(self, prompt, test_cases): ...
```

Extension patterns:

- **Domain-calibrated** — replace generic dimensions with domain rubrics (clinical precision, regulatory completeness, citation accuracy)
- **Golden-set** — compare against a curated reference set using semantic similarity rather than LLM judgment
- **Multi-judge** — call two models as judges; resolve disagreements by majority or confidence weighting
- **Regression** — store scores per run; flag when a prompt change drops any dimension by more than N points

All extend BasePromptEvaluator, implement three abstract methods, and register with EvaluationFactory. No changes to callers.

11.8 Testing Prompt Evaluators

```
from unittest.mock import patch, MagicMock
from k9_aif_abb.k9_agents.evaluation.k9_prompt_evaluator import K9PromptEvaluator
from k9_aif_abb.k9_agents.evaluation.models.evaluation import PromptTestCase
from k9_aif_abb.k9_inference.models.inference_response import InferenceResponse

def _make_evaluator(config=None):
    return K9PromptEvaluator(
        config=config or {"pass_threshold": 70.0, "judge_model": "reasoning"}
    )

def test_evaluate_returns_grade():
    judge_json = (
        '{"correctness":{"score":90,"rationale":"correct"},'
        '"completeness":{"score":85,"rationale":"complete"},'
        '"format_compliance":{"score":80,"rationale":"formatted"},'
        '"clarity":{"score":88,"rationale":"clear"},'
        '"relevance":{"score":92,"rationale":"relevant"},'
        '"overall_rationale":"Good response."}'
    )
    mock_resp = MagicMock(spec=InferenceResponse)
    mock_resp.output = judge_json

    with patch(
        "k9_aif_abb.k9_agents.evaluation.k9_prompt_evaluator.llm_invoke",
        return_value=mock_resp,
    ):
        result = _make_evaluator().evaluate(
            prompt="Summarize the following:",
            input_data={"text": "K9-AIF is..."},
            actual_output="K9-AIF is a framework...",
            expected="A concise summary of K9-AIF",
        )

    assert result.grade in ("A", "B", "C", "D", "F")
    assert result.verdict in ("PASS", "FAIL")
    assert 0 <= result.score <= 100

def test_run_suite_pass_rate():
    judge_json = (
        '{"correctness":{"score":80,"rationale":"ok"},'
        '"completeness":{"score":80,"rationale":"ok"},'
        '"format_compliance":{"score":80,"rationale":"ok"},'
        '"clarity":{"score":80,"rationale":"ok"},'
    )
```

```

        "relevance":{"score":80,"rationale":"ok"},'
        "overall_rationale":"Acceptable."}'
    )
    mock_resp = MagicMock(spec=InferenceResponse)
    mock_resp.output = judge_json

    test_cases = [
        PromptTestCase(input_data={"x": 1}, expected="result 1"),
        PromptTestCase(input_data={"x": 2}, expected="result 2"),
    ]

    with patch(
        "k9_aif_abb.k9_agents.evaluation.k9_prompt_evaluator.llm_invoke",
        return_value=mock_resp,
    ):
        suite = _make_evaluator().run_suite("Test prompt", test_cases)

    assert suite.total == 2
    assert suite.pass_rate >= 0.0

```

Test coverage requirements:

- Grade boundary conditions (A/B/C/D/F thresholds)
- Weighted composite arithmetic
- PASS/FAIL threshold behaviour
- Custom threshold override
- Malformed judge JSON fallback — score=50, no crash
- compare() winner detection and tie detection
- run_suite() aggregation for all-pass and all-fail states

11.9 K9Chat Integration

K9Chat includes an evaluation toggle in the topbar for development-time prompt grading. When enabled:

- A grade pill (A-F) appears beneath each assistant message
- Hover to see the composite score, verdict, and judge rationale per dimension
- Toggle the provider model in settings to observe the grade change — that delta is signal for which model executes the authored prompt best

This is a **development tool, not a production feature**. Enable it during prompt authoring; disable it in production deployments.

evaluation:

```

enabled: false      # toggle via /chat/evaluation/toggle endpoint
provider: k9
pass_threshold: 70
judge_model: reasoning

```

11.10 When to Use Prompt Evaluation

Use this pattern when:

- An agent's system prompt or instruction set has been authored and you need to verify it performs as intended before deploying
- You are comparing two prompt variants and need a data-backed decision on which to adopt
- A model upgrade or provider change occurred and you need to verify existing prompts still perform acceptably
- A new guided-flow step is being designed and you want to gate promotion on a minimum grade

Do not use it for:

- **Runtime quality enforcement** — that is K9ValidationLoopAgent's role
 - **User input validation** — prompt evaluation targets authored prompts, not user messages
 - **Agent regression testing** — domain-specific behaviour belongs in the domain test suite
-

12. Model Routing and Inference

12.1 The Inference Pipeline

Agents must never call LLM providers directly. All LLM invocations go through `llm_invoke()`:

```
llm_invoke(config, InferenceRequest)
→ ModelRouterFactory.get_router(config)      # cached router instance
→ K9ModelRouter.route(request)                # weighted scoring
→ catalog.get_model(best_alias)               # model metadata lookup
→ LLMFactory.get(llm_ref)                    # cached LLM instance
→ OllamaLLM.invoke(prompt)                   # actual inference call
→ RouteDecision + scores persisted to RoutingStateStore
→ InferenceResponse returned
```

12.2 `llm_invoke`

`k9_utils/llm_invoke.py` — the canonical LLM call path:

```
from k9_aif_abb.k9_utils.llm_invoke import llm_invoke
from k9_aif_abb.k9_inference.models.inference_request import InferenceRequest

req = InferenceRequest(
    prompt="Evaluate this claim for coverage...",
    task_type="reasoning",           # +3 scoring bonus if model has this capability
    sensitivity="confidential",      # +2 scoring bonus if model supports "confidential"
    latency_budget="interactive",    # +2 scoring bonus if model's latency_tier matches
```



```

        cost_profile="standard",      # +2 scoring bonus if model's cost_tier matches
        metadata={"agent": "ClaimsAdjudicationAgent", "claim_id": "C-001"},
    )

    resp = llm_invoke(self.config, req)
    resp.output      # LLM text output
    resp.model_alias # which model was selected
    resp.provider    # "ollama"
    resp.latency_ms  # round-trip latency

```

If the LLM backend is unreachable, `llm_invoke` raises `RuntimeError`. Always handle it:

```

try:
    resp = llm_invoke(self.config, req)
except RuntimeError as exc:
    self.logger.error("[%s] LLM unavailable: %s", self.layer, exc)
    return {"agent": self.layer, "output": "[WARN] LLM unavailable", "confidence": 0.0}

```

12.3 InferenceRequest

```

class InferenceRequest(BaseModel):
    prompt: str
    system_prompt: Optional[str] = None
    task_type: Optional[str] = None          # drives K9ModelRouter scoring
    max_tokens: Optional[int] = None
    temperature: Optional[float] = None
    sensitivity: Optional[str] = None        # "confidential" → +2 routing bonus
    latency_budget: Optional[str] = None     # "realtime" | "interactive" | "batch"
    cost_profile: Optional[str] = None       # "minimal" | "standard" | "premium"
    metadata: Optional[Dict[str, Any]] = None

```

12.4 K9ModelRouter Scoring

`K9ModelRouter` selects the best model from the catalog using weighted scoring:

Signal	Condition	Points
Task type match	<code>request.task_type</code> is in <code>model's capabilities[]</code>	+3
Confidential sensitivity	<code>request.sensitivity == "confidential"</code> AND <code>model</code> has "confidential" capability	+2
Latency budget match	<code>request.latency_budget</code> matches <code>model's latency_tier</code>	+2

Signal	Condition	Points
Cost profile match	request.cost_profile matches model's cost_tier	+2

The model with the highest score is selected. Falls back to `default_model` when no model scores above zero.

12.5 Model Catalog Configuration

```
inference:
  router:
    type: k9_model_router
    default_model: general
    persistence:
      enabled: true
      provider: sqlite
      sqlite:
        db_path: "./runtime/k9_model_router.db"

  llm_factory:
    base_url: "${OLLAMA_BASE_URL:-http://localhost:11434}"
    models:
      general: "llama3.2:1b"
      reasoning: "granite3-dense:2b"
      enterprise: "llama3.1:latest"

  models:
    general:
      provider: ollama
      llm_ref: general
      capabilities: [general, chat, summarization]
      latency_tier: realtime
      cost_tier: minimal

    reasoning:
      provider: ollama
      llm_ref: reasoning
      capabilities: [reasoning, analysis, extraction]
      latency_tier: interactive
      cost_tier: standard

    enterprise:
      provider: ollama
      llm_ref: enterprise
      capabilities: [enterprise, confidential, reasoning]
```

```
latency_tier: batch
cost_tier: premium
```

12.6 Custom Model Router

To replace K9ModelRouter with domain-specific routing logic:

```
from k9_aif_abb.k9_inference.routers.base_model_router import BaseModelRouter
from k9_aif_abb.k9_inference.models.inference_request import InferenceRequest
from k9_aif_abb.k9_inference.models.route_decision import RouteDecision

class ComplianceAwareRouter(BaseModelRouter):

    def route(self, request: InferenceRequest) -> RouteDecision:
        # Route sensitive data to the on-premise model only
        if request.sensitivity == "confidential":
            return RouteDecision(model_alias="enterprise", rationale="confidential data")
        if request.task_type == "reasoning":
            return RouteDecision(model_alias="reasoning")
        return RouteDecision(model_alias="general")

    def invoke(self, request: InferenceRequest):
        decision = self.route(request)
        llm = self._get_llm(decision.model_alias)
        return llm.invoke(request.prompt)

    async def ainvoke(self, request: InferenceRequest):
        decision = self.route(request)
        llm = self._get_llm(decision.model_alias)
        return await llm.ainvoke(request.prompt)
```

Register it in config.yaml by setting inference.router.type: compliance_aware_router.

12.7 Routing Decision Persistence

K9ModelRouter persists every routing decision to the RoutingStateStore. This provides:

- Complete audit trail of which model handled each request
- Session-level model affinity tracking
- Foundation for model performance analytics
- Support for compliance reporting

The state store uses SQLite by default and PostgreSQL when configured. Tables: sessions, session_turns, routing_decisions, context_artifacts.

12.8 Why Direct Model Calls Must Be Avoided

Concern	Direct Call	Via llm_invoke
Model selection	Hardcoded	Configuration-driven
Audit trail	None	Full routing decision log
Governance	None	Pre/post hooks available
Error handling	Custom per agent	Consistent across framework
Model swap	Requires code change	Config-only change
Observability	None	LLMCall trace events

13. Governance and Zero Trust

13.1 The Governance Pipeline

Every K9-AIF component receives a governance pipeline at construction via `require_governance()`. The pipeline provides two lifecycle hooks:

```
class BaseGovernance(ABC):
```

```
    @abstractmethod
    async def pre_process(self, payload: Dict[str, Any], ctx=None) -> Dict[str, Any]:
        """Apply governance BEFORE payload leaves the component."""

    @abstractmethod
    async def post_process(self, payload: Dict[str, Any], ctx=None) -> Dict[str, Any]:
        """Apply governance AFTER payload is received."""
```

13.2 require_governance()

`k9_core/governance/pipeline.py` — governance resolution at component init:

```
def require_governance(governance, env: str | None = None) -> Any:
    """
    If governance is provided → use it.
    If None:
        - development / test → WARNING log, return NoopGovernance (permitted)
        - production / staging → ERROR log, return NoopGovernance (dangerous – enforce_
    """
```

The `K9_ENV` environment variable controls this behavior. Always set it appropriately:

```
export K9_ENV=development    # local development
export K9_ENV=test           # CI/testing
export K9_ENV=staging        # pre-production
export K9_ENV=production     # production
```

13.3 NoopGovernance

NoopGovernance is a passthrough — it returns the payload unchanged. It is valid only in development and test environments. In staging or production, any component that calls `enforce_governance()` will raise `PermissionError` if NoopGovernance is active.

13.4 ProfanityGovernance OOB

`k9_governance/profanity_governance.py` provides a content-filtering governance implementation using an LLM guardian model:

```
# config/governance.yaml
governance:
  enabled: true
  policies:
    - type: LLMGovernance
      enabled: true
      apply_pre: true
      apply_post: true
      provider: "ollama"
      model: "granite-guardian"
      max_tokens: 64
      prompt_template: |
        You are a safety guardrail. Review this text:
        "{text}"
        Respond SAFE or BLOCK.
```

13.5 Zero Trust Security Layer

`k9_security/zero_trust/` implements a runtime zero-trust execution model. It is optional and enabled per-component via `enable_zero_trust: true` in config or at construction.

The zero-trust layer operates on an `ExecutionContext` that captures:

```
@dataclass
class ExecutionContext:
    request_id: str
    session_id: Optional[str]
    workflow_id: Optional[str]
    source_type: str # "router" | "orchestrator" | "agent"
    action_type: str # "route" | "execute_flow" | "validate"
    identity: IdentityContext # principal_id, principal_type, roles, tenant_id
    attributes: AttributeContext # data_sensitivity, environment, trust_zone
    destination: DestinationContext # destination_type, name, uri, is_external
    payload: Dict[str, Any]
```

The decision flow:

```
ZeroTrustGuard.evaluate(context) → ZeroTrustDecision (allowed, risk_score, obligation)
PolicyEnforcer.enforce(ctx, decision) → final ZeroTrustDecision
```

Results are returned as:

```
{
    "allowed": True | False,
    "decision": "ALLOWED" | "DENIED" | "BYPASSED",
    "reason": str,
    "risk_score": float,
    "obligations": list,
    "payload": dict,
}
```

13.6 Governance vs Zero Trust

These two mechanisms are architecturally distinct:

Aspect	Governance	Zero Trust
Concern	Policy intent (what is allowed)	Execution control (is this execution context permitted)
Scope	Payload content	Identity, context, risk
Hooks	pre_process / post_process	Single evaluate + enforce
Default	NoopGovernance	Disabled unless enable_zero_trust: true
ABB	BaseGovernance	BaseZeroTrustGuard, BasePolicyEnforcer

13.7 Governance Enforcement Patterns

Pattern 1: Enforce governance is configured (fail-fast in production)

```
def execute(self, payload):
    try:
        self.enforce_governance()
    except PermissionError as exc:
        return {"agent": self.layer, "output": "[WARN] governance not configured"}
```

Pattern 2: Apply content governance hooks

```
import asyncio

payload = asyncio.get_event_loop().run_until_complete(
    self.apply_pre_governance(payload)
)
# ... processing ...
result = asyncio.get_event_loop().run_until_complete(
    self.apply_post_governance(result)
)
```

Pattern 3: Zero-trust evaluation in orchestrator

```

zt_result = self.apply_zero_trust(payload)
if not zt_result["allowed"]:
    return {"status": "denied", "reason": zt_result["reason"]}

```

13.8 Why Governance Must Not Be Bypassed

Governance bypasses in production create invisible risks. Payloads that bypass governance may:

- Contain injected instructions that manipulate LLM behavior
- Leak sensitive data in model outputs
- Produce outputs that violate regulatory requirements
- Leave no audit trail for compliance reporting

The framework makes bypassing governance explicit and traceable. NoopGovernance in production raises a `PermissionError` rather than silently passing — this is intentional.

14. Messaging, Events, and Telemetry

14.1 `publish_event()`

`publish_event()` is defined on `BaseAgent` and routes events to the wired monitor and message bus:

```

def publish_event(self, event: Dict[str, Any]):
    if self.message_bus:
        self.message_bus.publish(event)  # → Kafka / Redpanda if wired
    if self.monitor:
        self.monitor.record_event(event)  # → Prometheus / OTEL / console
    self.logger.info(f"[{self.layer}] Event published: {event}")

```

In standard K9-AIF solutions, agents are constructed without a `message_bus`. Their events reach the monitor and logger. Only the Router and Orchestrator are wired with a message bus for Kafka integration.

14.2 Message Bus Abstraction

The message bus abstraction is `K9EventBus` (`k9_core/messaging/k9_event_bus.py`), which provides a local publish/subscribe mechanism. For Kafka/Redpanda integration, `RedpandaStreamProvider` (`k9_core/streaming/redpanda_provider.py`) implements the `BaseStreamProvider` contract.

14.3 Framework Event Taxonomy

Component	Event Type	Description
Router	<code>EventRouted</code>	Routing decision made

Component	Event Type	Description
Orchestrator	status events via publish_status()	Flow lifecycle
BaseValidationLoopAgent	loop_started, loop_continued, loop_finalized, loop_escalated, loop_failed	Loop lifecycle
BaseValidationLoopAgent	hypothesis_generated, validation_tool_invoked, observation_evaluated	Per-iteration
BaseCriticActorAgent	loop_started, draft_generated, draft_refined, critique_produced, loop_accepted, loop_rejected, loop_escalated, loop_failed	Round lifecycle
llm_invoke	LLMCall	Per inference call (via trace callback)

14.4 LLM Trace Callback

```
from k9_aif_abb.k9_utils.llm_invoke import register_trace_callback
```

```
def my_trace_handler(event: Dict[str, Any]) -> None:
```

```
    # event: {type, agent, task_type, model, provider, latency_ms, tokens_in, tokens_out}
    metrics_client.record(event)
```

```
register_trace_callback(my_trace_handler)
```

14.5 BaseMonitor

k9_core/monitoring/base_monitoring.py

```
class BaseMonitor(ABC):
```

```
    @abstractmethod
```

```
    def emit_metric(self, name: str, value: float, tags: Optional[Dict] = None) -> None:
        """Emit numeric metric."""
```

```
    @abstractmethod
```

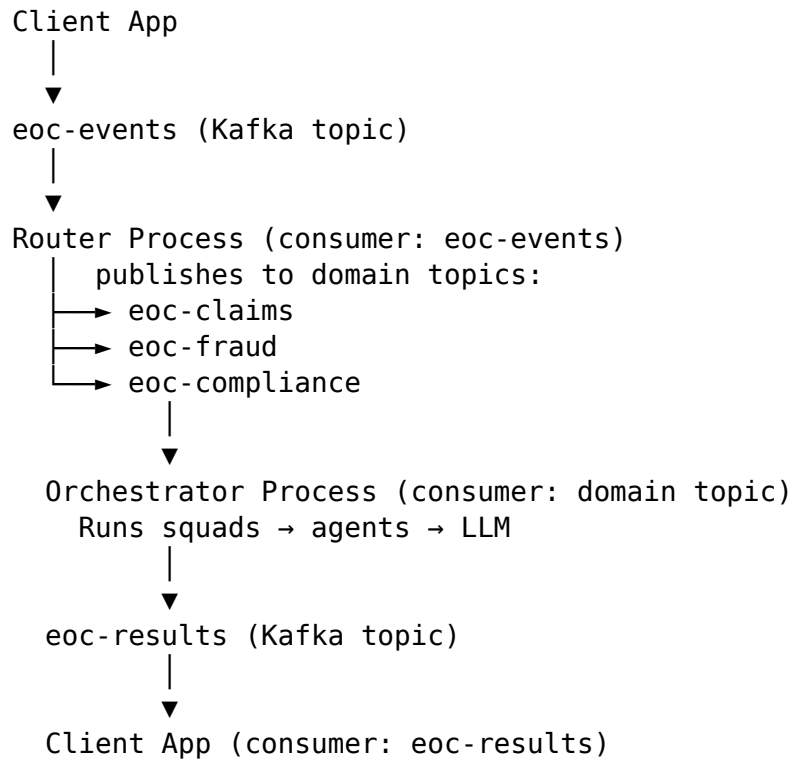
```
    def observe(self, event: str, meta: Optional[Dict] = None) -> None:
        """Record structured observation."""
```

OOB implementations: ConsoleMonitor, PrometheusMonitor, OTELMonitor, CloudWatchMonitor, LoggerMonitor.

The monitor is provisioned via MonitorFactory:

```
monitors:
  - name: logger
    class: k9_aif_abb.k9_core.monitoring.logger_monitor.LoggerMonitor
    kwargs: {}
```

14.6 Kafka Topology in Production



Configuration:

```
messaging:
  backend: redpanda
  broker_url: "${KAFKA_BROKER:-localhost:9092}"
  topic: k9aif-events
  group_id: k9aif-core
  client_id: k9aif-console
  security_protocol: PLAINTEXT
  auto_create: true
```

15. Persistence and Graph Integration

15.1 BasePersistence

k9_core/persistence/base_persistence.py — state storage contract:

```

class BasePersistence(ABC):

    @abstractmethod
    def save_state(self, key: str, state: Dict[str, Any]) -> None: ...

    @abstractmethod
    def load_state(self, key: str) -> Optional[Dict[str, Any]]: ...

    @abstractmethod
    def update_state(self, key: str, state: Dict[str, Any]) -> None: ...

    @abstractmethod
    def delete_state(self, key: str) -> None: ...

```

OOB implementations: SQLitePersistence, ChromaDBPersistence, MemoryPersistence.

15.2 BaseStorage

k9_core/storage/base_storage.py — file/object storage contract:

```

class BaseStorage(ABC):

    @abstractmethod
    def save(self, key: str, data: Any) -> None: ...

    @abstractmethod
    def load(self, key: str) -> Any: ...

    @abstractmethod
    def delete(self, key: str) -> None: ...

    @abstractmethod
    def list_keys(self, prefix: Optional[str] = None) -> list[str]: ...

```

OOB implementations: FileStorage, ObjectStorage, SQLiteDatabaseStorage, PostgresDatabaseStorage.

15.3 RoutingStateStore

k9_storage/routing_state_store.py — K9ModelRouter decision tracking:

Four tables managed automatically: - sessions — session metadata - session_turns — conversational turns - routing_decisions — model selection + scores + rationale - context_artifacts — metadata cache per turn

This store provides the data foundation for model performance analytics, governance reporting, and compliance audit trails.

15.4 PostgreSQL Integration

```
postgres:
  host: "${POSTGRES_HOST:-localhost}"
  port: 5432
  user: "postgres"
  password: "postgres"
  database: "k9aif"
  schema: "k9aif"           # all tables created under this schema
```

PostgresDatabaseStorage sets `search_path` and `MetaData(schema=...)` from `postgres.schema`. The schema value must match the PostgreSQL schema where tables reside.

15.5 Neo4j / Architecture Graph

K9-AIF is aligned with the <https://graph.k9x.ai> architecture graph, which represents the K9-AIF component topology as a property graph in Neo4j.

The architecture graph is intended for:

- Visualizing the ABB/SBB component hierarchy
- Tracking which SBBs realize which ABBs
- Documenting solution topology
- Supporting lineage queries (which agents participate in which workflows)

Connection details for the shared Neo4j instance:

```
bolt://${NEO4J_HOST:-localhost}:7687
```

Solution teams can write agent and squad registration events to Neo4j to maintain a live topology view of their solution.

15.6 Persistence Factory

```
from k9_aif_abb.k9_factories.persistence_factory import PersistenceFactory

persistence = PersistenceFactory.create(config)
# Creates SQLitePersistence, PostgresDatabaseStorage, or MemoryPersistence
# based on config.persistence[] entries
```

16. Configuration Standards

16.1 YAML Structure

K9-AIF uses YAML for all component configuration. The config hierarchy:

```
config/
├── config.yaml           # global infrastructure config (inference, messaging, postgres)
├── squads.yaml           # squad definitions and flow
```

```
└─ agents/yaml/          # one YAML file per agent
└─ governance.yaml      # governance policy definitions
```

16.2 Global config.yaml Sections

```
# Infrastructure – always in global config
inference:
  router: {...}
  llm_factory: {...}
  models: {...}

messaging:
  backend: redpanda
  broker_url: ...

postgres:
  host: ...
  schema: ...

governance:
  enabled: true
  policy_path: ...

# Demo / testing defaults
demo:
  default_intent: support
  query: ...
```

16.3 Agent Config Merging

When `agent_loader.merge_with_global(agent_name, global_config)` is called:

1. Global config.yaml is loaded
2. Agent agent_name.yaml is loaded
3. Agent YAML values override global values on key collision
4. The merged dict is passed as `agent.config`

```
# In agent code:
self.config.get("role")      # from agent YAML
self.config.get("model")     # from agent YAML
self.config.get("inference") # from global config.yaml
self.config.get("postgres")  # from global config.yaml
```

16.4 Naming Standards

Component	Naming Convention	Example
Agent class	PascalCase + “Agent”	ClaimsTriageAgent

Component	Naming Convention	Example
Agent YAML file	snake_case	claims_triage_agent.yaml
Squad ID	PascalCase + "Squad"	ClaimsProcessingSquad
Orchestrator class	PascalCase + "Orchestrator"	ClaimsOrchestrator
Router class	PascalCase + "Router"	E0CRouter
layer attribute	"<Solution> <Class> SBB"	"E0C ClaimsTriageAgent SBB"
Model alias	snake_case	reasoning, fast_local, enterprise
Event type	PascalCase	ClaimsTriageCompleted

16.5 Squad YAML Format

```
squads:
  SquadId:                                # squad ID – used by SquadLoader.load_one()
  description: "What this squad does."
  agents:
    - AgentOne                            # registered names
    - AgentTwo
    - AgentThree
  flow:
    - agent: AgentOne                      # MUST be a dict with 'agent:' key
      result_key: agent_one                # key in accumulated context
    - agent: AgentTwo
      result_key: agent_two
    when:                                  # conditional step
      key: "agent_one.approved"            # key in accumulated context
      eq: true
    - agent: AgentThree
      result_key: agent_three
    context:                               # static overrides merged into step input
      mode: strict
```

Critical: Flow steps must be dicts with an `agent:` key. Plain strings raise `ValueError` at runtime.

16.6 Environment Configuration

```
# Required for governance enforcement behavior
export K9_ENV=development # or test | staging | production

# Infrastructure (typically set in container environment or .env)
export OLLAMA_BASE_URL=http://localhost:11434
export POSTGRES_HOST=localhost
export KAFKA_BROKER=localhost:9092
```

17. Testing Standards

17.1 Testing Philosophy

K9-AIF tests are divided into two categories:

- **Framework stability tests** — test ABB contracts; no external services required
- **Domain behavior tests** — test SBB implementations; may mock LLM

Framework tests live in `k9_aif_abb/tests/`. Domain tests live alongside the SBB code.

17.2 Testing Agents Offline

Always mock `llm_invoke` — never call a real LLM in unit tests:

```
from unittest.mock import patch, MagicMock
from k9_aif_abb.k9_inference.models.inference_response import InferenceResponse
from examples.my_app.agents.src.my_agent import MyAgent
```

```
class _TestGovernance:
    """Minimal concrete governance – define inline in each test file."""
    def pre_process(self, payload: dict, ctx=None) -> dict:
        return payload
    def post_process(self, payload: dict, ctx=None) -> dict:
        return payload
```

```
def _make_agent(config=None):
    return MyAgent(config=config or {}, governance=_TestGovernance())
```

```
def test_execute_returns_expected_output():
    mock_resp = MagicMock(spec=InferenceResponse)
    mock_resp.output = '{"decision": "approved", "confidence": 0.95}'
    mock_resp.model_alias = "reasoning"
    mock_resp.provider = "ollama"
```

```
    with patch("examples.my_app.agents.src.my_agent.llm_invoke", return_value=mock_resp):
        agent = _make_agent()
        result = agent.execute({"claim_id": "C001", "amount": 5000})
```

```
    assert "output" in result
    assert result["model_used"] == "reasoning"
```

```
def test_execute_handles_llm_error():
    with patch("examples.my_app.agents.src.my_agent.llm_invoke",
               side_effect=RuntimeError("LLM backend unavailable")):
        agent = _make_agent()
```

```

        result = agent.execute({"claim_id": "C001"})

    assert "[WARN]" in result.get("output", "")

```

17.3 Testing Validation Loop Agents

```

from k9_aif_abb.k9_agents.validation.models.validation_loop import (
    ValidationDisposition, ValidationLoopContext
)

class _FixedLoopAgent(BaseValidationLoopAgent):
    """Test double – deterministic, no external dependencies."""
    def generate_hypothesis(self, ctx): return {"query": "test"}
    def run_validation(self, h, ctx): return {"match": True, "score": 0.9}
    def evaluate_observation(self, r, ctx): return {"confidence": 0.9, "covered": r["ma
    def should_continue(self, obs, ctx):
        if obs["confidence"] >= 0.8:
            return ValidationDisposition.FINALIZE
        return ValidationDisposition.CONTINUE
    def finalize(self, ctx):
        from k9_aif_abb.k9_agents.validation.models.validation_loop import ValidationLo
        return ValidationLoopResult(
            disposition=ValidationDisposition.FINALIZE,
            output={"decision": "approved"},
            steps=ctx.steps,
            iterations=ctx.iteration,
            final_confidence=0.9,
        )

def test_validation_loop_finalizes():
    agent = _FixedLoopAgent(config={}, governance=_TestGovernance())
    result = agent.execute({"claim_id": "C001"})
    assert result["disposition"] == "finalize"
    assert result["final_confidence"] == 0.9
    assert result["iterations"] >= 1

```

15.4 Testing Critic-Actor Agents

```

class _FixedCriticAgent(BaseCriticActorAgent):
    """Test double – accepts on second round."""
    def generate(self, ctx): return "initial draft"
    def critique(self, draft, ctx):
        if ctx.round == 1:
            return {"accepted": False, "score": 0.5, "issues": ["too vague"]}
        return {"accepted": True, "score": 0.9, "issues": []}
    def refine(self, draft, feedback, ctx): return "refined draft"
    def should_accept(self, feedback, ctx):

```

```

        if feedback["accepted"]:
            return CriticActorDisposition.ACCEPTED
        return CriticActorDisposition.REJECTED
    def finalize(self, ctx):
        from k9_aif_abb.k9_agents.critic_actor.models.critic_actor import CriticActorResult
        return CriticActorResult(
            disposition=CriticActorDisposition.ACCEPTED,
            output={"final_draft": ctx.steps[-1].draft},
            steps=ctx.steps,
            rounds=ctx.round,
            final_score=0.9,
        )

def test_critic_actor_converges():
    agent = _FixedCriticAgent(config={}, governance=_TestGovernance())
    result = agent.execute({"task": "write report"})
    assert result["disposition"] == "accepted"
    assert result["rounds"] == 2

```

15.5 ABB Testing Principles

Principle	Rationale
Test the contract, not the implementation	ABB tests verify the loop skeleton, not domain logic
Never import production governance in tests	Define <code>_TestGovernance</code> inline in every test file
Mock at the boundary, not inside components	Mock <code>llm_invoke</code> , not internal LLM calls
Use MockLLM from <code>k9_core/inference/mock_llm.py</code>	Deterministic responses without external deps
Test all dispositions (FINALIZE, ESCALATE, FAIL)	Verify terminal paths, not just the happy path

15.6 Running the Test Suite

```

# Activate the virtual environment
source k9_aif_abb/.venv/bin/activate

# Framework stability tests (no external services)
cd k9_aif_abb
pytest tests/test_framework.py -v
pytest tests/test_intelligent_model_router.py -v

# All framework tests
pytest tests/ -v

```



```
# Single test file
pytest tests/test_k9_validation_loop_agent.py -v
```

18. Developer Workflow

18.1 Environment Setup

```
# Clone the repository
git clone https://github.com/k9aif/k9-aif-framework.git
cd k9-aif-framework

# Create virtual environment
python3.11 -m venv .venv
source .venv/bin/activate

# Install dependencies
pip install -r requirements.txt

# Set environment
export K9_ENV=development
```

18.2 Adding a New ABB

1. Identify the capability gap — only add an ABB if no existing ABB covers the need
2. Create the abstract class in the appropriate k9_core/ sub-package
3. Define the minimal contract: abstract methods only
4. Add governance hook wiring if the component handles payloads
5. Add layer class attribute
6. Export from the package `__init__.py`
7. Write framework stability tests
8. Update the package docstring in `__init__.py`

18.3 Adding a New SBB

```
# Use the generator for complete scaffold
./k9_generator.sh preview MyNewApp    # preview without writing
./k9_generator.sh run MyNewApp        # write scaffold to k9_projects/MyNewApp/
```

The generator creates: - Agent stubs extending BaseAgent - Squad YAML configuration
- Orchestrator stub extending BaseOrchestrator - Router stub extending BaseRouter -
Config YAML - Test stubs

After generation, flesh out the stubs using the EOC example as reference.

18.4 Development Checklist

Before submitting any agent or orchestrator:

- [] Agent extends BaseAgent (or appropriate iterative pattern)
- [] layer attribute is set (e.g., "EOC MyAgent SBB")
- [] execute() handles RuntimeError from llm_invoke
- [] All LLM calls go through llm_invoke(), not direct Ollama calls
- [] Agent YAML has no squad or routing fields
- [] Squad YAML has no orchestrator field
- [] Flow steps are dicts with 'agent:' key (not plain strings)
- [] Tests mock llm_invoke, not internal LLM classes
- [] _TestGovernance defined inline in test file
- [] publish_event() called for significant outcomes
- [] enforce_governance() called if agent handles regulated data
- [] K9_ENV set appropriately for test environment

18.5 Common Commands

Run framework tests

```
pytest k9_aif_abb/tests/test_framework.py -v
```

Run example applications

```
./run_k9chat.sh
```

```
./run_acme_support_center.sh
```

Smoke tests

```
bash test_model_router.sh
```

```
bash test_squads.sh
```

Generator

```
./k9_generator.sh preview <AppName>
```

```
./k9_generator.sh run <AppName>
```

Build and run EOC (RHEL/Podman)

```
bash build.sh
```

```
bash run_eoc_pod.sh
```

19. K9X Studio Integration

19.1 What Studio Does

K9X Studio is a visual architecture designer for K9-AIF systems. Architects design on a drag-and-drop canvas; developers receive a framework-compliant scaffold.

Install and run:

```
pip install k9x
```

```
k9x studio
```

Opens at <http://localhost:9494>.

19.2 Using Studio

1. **Start from a template** or drag components from the palette
2. **Palette components:** Router (indigo), Orchestrator (purple), Squad (blue), Agent (green), Validation Loop (amber), Critic-Actor (red), Guard (slate), HIL Orchestrator (teal)
3. **Connect components** — Studio validates the hierarchy (Router → Orchestrator → Squad → Agent)
4. **Configure** each component in the Inspector panel — name, agent type, model, LLM provider
5. **Generate scaffold** — exports a K9-AIF project with all files, YAML configs, and agent stubs

19.3 From Specification

Upload a markdown spec, BPMN file, or process description. Studio reads it and suggests an architecture on the canvas. Adjust and generate.

19.4 Scaffold Output

```
my_project/
├── config/
│   ├── config.yaml
│   └── squads.yaml
├── agents/
│   ├── yaml/           # Agent YAML configs
│   └── src/             # Agent Python stubs
├── orchestrators/
├── main.py
├── run.sh
└── setup.sh
```

Every stub extends the correct ABB. Squad YAML is pre-wired. Configuration is populated from the canvas.

20. K9X Enterprise Continuum

20.1 What the Continuum Does

The K9X Enterprise Continuum is a governed catalog where validated SBBs and ABBs are published, discovered, and evolved. It implements the TOGAF Enterprise Continuum as live, API-first infrastructure.

20.2 Publishing an SBB

Step 1 — Validate compliance:

```
k9aif inspect
```

Checks that every agent extends the correct ABB, squad YAML is valid, and configuration is complete.

Step 2 — Publish:

```
k9aif publish --name MyAgent --kind Agent --abb-names BaseAgent \
  --domain insurance --project MyProject
```

The SBB is registered in the Continuum with metadata: ABB contract, domain tag, version, inspection status, author.

20.3 Tier Classification

SBBs are classified into four tiers based on their ABB implementations and domain tags:

Tier	Criteria
Foundation	Implements Foundation ABBs, no domain tag
Common Systems	Implements Common Systems ABBs (validation loop, planning loop, etc.)
Industry	Has a domain tag (insurance, healthcare, defense)
Organization-Specific	Enterprise-customized, deployed applications

20.4 Querying the Continuum

The Continuum is API-first. Studio queries it before generating scaffolds.

GET /api/sbbs	# list all SBBs
GET /api/sbbs?domain=insurance	# filter by domain
GET /api/abbs	# list all ABBs
GET /api/abbs/{id}/sbbs	# SBBs implementing this ABB

21. Human-in-the-Loop Integration

21.1 When to Use HIL

Use HIL when an agent cannot decide with sufficient confidence, or when policy requires human review. The decision of where HIL belongs is made at design time using the 4Ds framework:

- **Delegation** — fully automated, no HIL
- **Description** — needs clear spec, no HIL
- **Discernment** — agent confidence is uncertain → HIL
- **Diligence** — regulation requires human sign-off → HIL

21.2 Wiring a HIL Breakpoint

In your orchestrator, when the agent's confidence is below threshold:

```
if result.get("confidence", 1.0) < self.config.get("hil_threshold", 0.8):
    self.publish_event({
        "type": "hil.task.created",
        "correlation_id": payload.get("correlation_id"),
        "reply_to": "workflow.eoc.hil.response",
        "title": f"Review: {payload.get('claim_id')}",
        "description": "Agent confidence below threshold",
        "ttl_hours": 168,
        "ttl_action": "reject",
        "payload": result,
    })
    return {"status": "awaiting_hil", "correlation_id": payload.get("correlation_id")}
```

The orchestrator publishes and exits. No thread blocked. No resource consumed while waiting.

21.3 Consuming the HIL Response

The HILOrchestrator subscribes to the reply_to topic:

```
class EOCHILOrchestrator(BaseOrchestrator):
    def orchestrate(self, event: dict) -> dict:
        human_decision = event.get("decision")
        correlation_id = event.get("correlation_id")
        # Continue the pipeline with the human's decision in context
        return self.run_squad({"decision": human_decision, "correlation_id": correlation_id})
```

21.4 Studio Canvas

In K9X Studio, drop the HIL Orchestrator from the palette. It auto-creates a HILSquad and HILAgent, connected to the Kafka bus via a teal dotted line. No Router needed — the HIL Orchestrator is event-driven.

21.5 K9X HIL Platform

K9X HIL is a dedicated case management platform for human tasks:

```
pip install k9x-hil
```

- **Project → Application → Queue → Task** hierarchy
- **Role-based access** — Root, Project Admin, App Admin, Manager, HIL Operator
- **TTL enforcement** — tasks expire if humans don't act
- **PII awareness** — incoming message declares sensitive fields
- **Kafka-native** — works with any system that writes JSON to Kafka

22. Provider Adapter Pattern

22.1 The Three-Layer Structure

Every infrastructure concern in K9-AIF follows the same pattern:

1. **ABB Contract** — abstract base class defining the interface
2. **Concrete Adapter** — wraps a vendor SDK behind the ABB contract (lazy imports)
3. **Factory** — reads config, resolves the correct adapter, returns the uniform interface

22.2 Implemented Areas

Concern	ABB Contract	Default Adapter	Other Adapters	Factory	Config Key
Inference	BaseLLM	OllamaLLM	OpenAI, Anthropic, watsonx, Azure	LLMFactory	inference.llm_factory
Model Routing	BaseModelRouter	K9ModelRouter	Custom via config	ModelRouterFactory	inference.router
Secret Management	BaseSecretManager	K9SecretAdapter	Vault, AWS, IBM	SecretManagerFactory	secret.provider
Cache	BaseCache	InMemoryAdapter	Redis	CacheFactory	cache.provider

22.3 Adding a New Adapter

```
# k9_<concern>/adapters/my_adapter.py
```

```
class MyAdapter(BaseSecretManager):
    def __init__(self, config=None):
        self._config = config or {}
        self._client = None

    def _ensure_client(self):
        if self._client is not None:
            return
        try:
            import my_package
            self._client = my_package.Client(...)
        except ImportError as exc:
            raise RuntimeError("pip install my_package required") from exc

    def get(self, key: str) -> str:
        self._ensure_client()
        return self._client.get_secret(key)
```

Register with the factory:

```
SecretManagerFactory.register("my_provider", MyAdapter)
```

Or via config:

```
secrets:  
  provider: my_provider
```

22.4 Constraints

- Credentials NEVER in config.yaml — use environment variables
 - Optional packages: lazy import, raise RuntimeError with install hint
 - Factory create(config) always has a zero-config default
 - All adapter code is purely additive — no modification to existing classes
-

23. Using Claude Code with VSCode for K9-AIF Development

23.1 Overview

Claude Code is an AI-assisted coding assistant that integrates with VSCode and operates via the Claude Code CLI. K9-AIF uses Claude Code as a development accelerator — not as the architect. The human architect retains authority over all structural decisions.

This chapter documents lessons learned from using Claude Code throughout K9-AIF development and provides guidance for teams adopting it.

23.2 Recommended Developer Workflow

The effective K9-AIF + Claude Code workflow is:

1. **Architect designs the structure:** Identify which agents, squads, and orchestrators are needed. Make ABB vs iterative pattern decisions explicitly before writing code.
2. **Write the YAML first:** Agent YAML, squad YAML, and config YAML define the behavioral specification. Commit these before generating code.
3. **Use Claude Code for implementation:** With the YAML spec in place, Claude Code generates the Python implementations reliably.
4. **Review all generated code critically:** Claude Code does not always preserve framework boundaries. Review every generated file before committing.
5. **Run framework tests immediately:** The hooks in .claude/settings.json run test_framework.py automatically after every write — any regression is visible immediately.

23.3 CLAUDE.md Structure

The CLAUDE.md file is the primary mechanism for teaching Claude Code about the K9-AIF framework. It should contain:

- **Architecture overview:** ABB/SBB distinction, execution hierarchy, three-level decoupling
- **Key contracts:** `execute_flow()`, `execute()`, `route()` — with correct signatures
- **Decoupling rules:** Squad YAML has no orchestrator field; agent YAML has no squad field
- **LLM invocation path:** Why `llm_invoke()` is mandatory and how to use it
- **Governance rules:** `require_governance()`, `enforce_governance()`, environment behavior
- **Anti-patterns to avoid:** Direct Ollama calls, bypassing governance, agent-to-agent Kafka

The `CLAUDE.md` in this repository is authoritative. Read it before using Claude Code for any K9-AIF development.

23.4 Purpose of SKILLS.md

`SKILLS.md` provides step-by-step recipes that Claude Code can follow for common tasks:

- Skill 1: Add a new Agent
- Skill 2: LLM invocation pattern
- Skill 3: Custom Model Router
- Skill 4: Add a new Squad
- Skill 5: Governance enforcement
- Skill 6: Write agent tests
- Skill 7: Event publishing
- Skill 8: Add model routing signal
- Skill 9: Agent config merging
- Skill 10: Build a validation loop agent
- Skill 11: Provider Adapter Pattern (Secret Management, Cache, and future areas)
- Skill 12: Apply the 4Ds AI Fluency Framework to Agent Design
- Skill 13: Add a new LLM provider adapter (Claude, Bedrock, OpenAI, etc.)
- Skill 14: Chat session management (multi-turn conversation context)
- Skill 15: Streaming LLM responses (token-by-token output)

Reference these skills in prompts: “Follow Skill 1 to add a new `FraudAssessmentAgent`.”

23.5 Teaching Claude Code Framework Conventions

Claude Code learns framework conventions from `CLAUDE.md` context. For K9-AIF, the most critical conventions to reinforce are:

Inheritance patterns:

```
# Always inherit from the correct base
class MyAgent(BaseAgent):           # one-shot agents
class MyLoopAgent(K9ValidationLoopAgent): # iterative convergence
class MyRefinementAgent(K9CriticActorAgent): # actor-critic refinement
class MyOrchestrator(BaseOrchestrator):
```



```
class MyRouter(BaseRouter):
```

Never tell Claude Code to: - “Connect the agent to Kafka” — agents don’t own Kafka in standard K9-AIF - “Create a direct Ollama connection” — always use `llm_invoke()` - “Save the governance check for later” — governance must be wired at init

23.6 Common Mistakes AI Coding Assistants Make in Framework Development

Mistake	Correct Pattern
Calling <code>OllamaLLM</code> directly	Use <code>llm_invoke(self.config, InferenceRequest(...))</code>
Omitting the <code>layer</code> attribute	Always set <code>layer = "<App> <Name> SBB"</code>
Adding <code>squad: field</code> to agent YAML	Agent YAML has no <code>squad</code> reference
Adding <code>orchestrator: field</code> to squad YAML	Squad YAML has no <code>orchestrator</code> reference
Using <code>execute()</code> in a validation loop agent	Replace with the five loop methods
Importing <code>NoopGovernance</code> in test governance	Define <code>_TestGovernance</code> inline
Using plain strings in squad flow steps	Flow steps must be dicts with <code>agent: key</code>
Calling <code>asyncio.run()</code> inside an <code>async def</code>	Use <code>await</code> instead

23.7 Guiding Claude Code Toward Correct Patterns

Prompt patterns that work well:

"Add a new agent called `FraudAssessmentAgent` following Skill 1 in `SKILLS.md`. It should extend `K9ValidationLoopAgent` since it needs iterative confidence building. The validation tool is the `FraudRuleEngine` class in `examples/eoc/rules/fraud_rules.py`. Do not call `OllamaLLM` directly – use `llm_invoke()`."

"Write offline tests for `FraudDetectionAgent` following Skill 6 in `SKILLS.md`. Mock `llm_invoke` at the module level. Test the `FINALIZE` path, the `ESCALATE` path, and the LLM error handling path. Define `_TestGovernance` inline."

"The squad YAML for `ClaimsProcessingSquad` has plain strings in the flow list. Fix them to be dicts with an `'agent:'` key and appropriate `'result_key'` values."

Prompt patterns that cause problems:

- “Make the agent smarter” — too vague; causes hallucinated capabilities
- “Wire everything together” — causes boundary violations
- “Add persistence to the agent” — agents should not own persistence; orchestrators do
- “Generate the whole solution” — one-shot generation misses framework boundaries

23.8 Incremental Generation vs Large One-Shot Generation

Incremental is better. Generate one component at a time:

1. Agent YAML first
2. Agent Python class second
3. Tests third
4. Squad YAML fourth
5. Orchestrator fifth
6. Router sixth

One-shot generation of a complete solution reliably produces: - Agents that reference squads - Direct LLM calls - Missing governance wiring - Incorrect squad YAML format

Incremental generation with review between steps produces code that matches the framework contract.

23.9 Reviewing AI-Generated Code Critically

Review every generated file for:

1. **Inheritance:** Is the parent class correct? Is `K9ValidationLoopAgent` used where iterative convergence is needed?
2. **LLM calls:** Does `llm_invoke()` appear? No direct `OllamaLLM` calls?
3. **Governance:** Is `require_governance()` called at init? Is `enforce_governance()` called for regulated agents?
4. **Squad YAML:** Are flow steps dicts? No orchestrator reference?
5. **Agent YAML:** No squad reference? No routing fields?
6. **Tests:** Is `_TestGovernance` inline? Is `llm_invoke` mocked?
7. **Layer attribute:** Set to "`<App> <Name> SBB`"?
8. **Error handling:** Does `execute()` handle `RuntimeError` from `llm_invoke`?

23.10 Examples of Successful Claude Code Usage in K9-AIF

Validation Loop Pattern: Claude Code was used to generate the initial `BaseValidationLoopAgent` skeleton and `K9ValidationLoopAgent` implementation. The abstract method signatures, loop lifecycle, and telemetry event names were specified precisely in the prompt. The resulting code required minimal revision.

Critic-Actor Pattern: `BaseCriticActorAgent` and `K9CriticActorAgent` were similarly generated with Claude Code after `BaseValidationLoopAgent` was complete, using it as a structural template. The pattern was specified in terms of the loop skeleton, terminal dispositions, and config key names.

Offline Test Generation: Claude Code consistently produces accurate offline tests when the prompt specifies: - Which agent to test - That `llm_invoke` must be mocked - That `_TestGovernance` must be defined inline - Which paths to test (happy path, error path, escalation path)

OOB Implementations: Generating `K9ModelRouter` from `BaseModelRouter` worked well when the scoring logic was specified explicitly in the prompt, including the exact point values.

23.11 Maintaining Architecture Authority

AI coding assistants accelerate implementation. They do not make architectural decisions. The human architect must:

- Decide which agents are iterative vs one-shot **before** generating code
- Decide the squad flow structure **before** generating agents
- Decide where governance is required **before** generating orchestrators
- Review all generated code against the ABB contracts
- Not accept AI-suggested “improvements” that violate framework boundaries

A useful discipline: write the YAML specification files yourself, then let Claude Code generate the Python from them. The YAML captures your architectural intent; the Python is an implementation detail.

23.12 Recommended Tool Combination

Tool	Role
Claude Code	Python and YAML generation, test generation, refactoring
VSCode	Code editing, file navigation, integrated terminal
Git	Version control; commit after every verified working component
Local Ollama	LLM backend for development and testing
Remote Ollama	LLM backend for production deployment
Architecture graph	graph.k9x.ai — validate topology against intent

23.13 Practical Lessons Learned

1. **Specify the exact base class in every prompt.** Claude Code defaults to BaseAgent; remind it explicitly when a different base is needed.
2. **Specify what NOT to do.** “Do not call OllamaLLM directly” prevents a common mistake.
3. **Run tests immediately after generation.** The CI hooks are fast; don’t batch changes before testing.
4. **Keep prompts concrete, not abstract.** “Add a FraudDetectionAgent that extends K9ValidationLoopAgent and uses FraudRuleEngine as the validation tool” is better than “add a fraud agent.”
5. **Architecture first, code second.** If you are not clear on the architecture, Claude Code will make up a plausible one that may not match your intent.
6. **Treat generated code as a first draft.** Always read it. The quality of the review matters more than the speed of generation.

24. Author's Recommendations

24.1 Keep ABBs Small and Stable

Every addition to `k9_aif_abb/` is a commitment to all solutions built on the framework. Add an ABB only when you identify a genuine cross-solution contract that all implementations will need to realize. When in doubt, implement it in an SBB first and promote it to ABB after it has proven stable.

24.2 Do Not Overgeneralize Too Early

The temptation to build a maximally general framework is strong. Resist it. A framework that tries to handle every possible case handles none of them well. Start with concrete problems, extract the general pattern once you have three implementations, and promote it to an ABB only when the contract is stable.

24.3 Prefer Explicit Architecture Decisions

Implicit architecture decisions accumulate as technical debt. When you choose `BaseValidationLoopAgent` over `BaseAgent` for a specific agent, document that decision in the squad YAML description or agent YAML description. When you wire zero-trust, document why. Future developers — and AI coding assistants — need this context.

24.4 Keep Domain Logic in SBBs

The most common framework mistake is letting domain logic creep into ABBs. Review every change to `k9_aif_abb/` for domain knowledge. Prompts, business rules, domain constants, and connection strings belong in SBBs.

24.5 Preserve Framework Boundaries

The three-level decoupling — Router → Orchestrator → Squad → Agent — exists to make the system composable and testable. Breaking it (agents that publish to Kafka, squads that know their orchestrator, agents that call LLMs directly) produces systems that are harder to test, harder to govern, and harder to extend.

24.6 Do Not Bypass Governance or Telemetry Hooks

Governance and telemetry are structural requirements, not optional features. A system that bypasses them in production is not governed. The framework provides mechanisms to make bypassing visible (`NoopGovernance` raises in production); use them.

24.7 Favor Architecture Consistency Over Feature Explosion

A consistent, limited feature set is more valuable than a large, inconsistent one. K9-AIF covers the major patterns: one-shot agents, validation loops, actor-critic refinement, squad orchestration, intent routing, model routing, governance, and zero-trust. These

are sufficient for most enterprise AI workflows. Resist adding patterns that do not fit this vocabulary.

24.8 Prefer Composable Runtime Patterns

Design agents and squads to be composable — an agent in one squad should work in a different squad without modification. An orchestrator should be invocable by a different router without modification. Composability is a direct result of respecting the decoupling rules.

25. Patterns Reference

25.1 Classic Software Patterns in K9-AIF

K9-AIF's architecture draws on well-established software design patterns. Understanding these patterns helps explain why the framework is structured as it is.

For the complete patterns library, see <https://github.com/k9aif/k9aif-patterns>.

25.2 Template Method

Where used: BaseValidationLoopAgent, BaseCriticActorAgent

The Template Method pattern defines the skeleton of an algorithm in a base class and defers specific steps to subclasses. The loop in `execute()` is the template; `generate_hypothesis()`, `run_validation()`, etc. are the deferred steps.

```
# Template (fixed in BaseValidationLoopAgent)
def execute(self, payload):
    while not terminal:
        hypothesis = self.generate_hypothesis(loop_ctx)      # deferred
        result = self.run_validation(hypothesis, loop_ctx)   # deferred
        obs = self.evaluate_observation(result, loop_ctx)    # deferred
        disposition = self.should_continue(obs, loop_ctx)    # deferred
        if disposition == FINALIZE:
            return self.finalize(loop_ctx)                   # deferred
```

25.3 Strategy

Where used: BaseModelRouter / K9ModelRouter

The Strategy pattern defines a family of algorithms, encapsulates each, and makes them interchangeable. Model routing strategies — weighted scoring, compliance-aware, cost-optimized — are interchangeable via config without changing agent code.

25.4 Factory

Where used: LLMFactory, ModelRouterFactory, PersistenceFactory, MonitorFactory

The Factory pattern centralizes object creation and decouples callers from concrete classes. All major framework components are provisioned through factories, never instantiated directly in application code.

Always this:

```
router = ModelRouterFactory.get_router(config)
```

Never this:

```
router = K9ModelRouter(catalog=..., config=..., state_store=...)
```

25.5 Chain of Responsibility

Where used: Handler, AgentHandler in k9_core/orchestration/base_handler.py

The Chain of Responsibility pattern passes a request along a chain of handlers until one handles it. AgentHandler wraps agents in a CoR chain, allowing pre/post processing at each step without modifying the agents themselves.

25.6 Observer / Eventing

Where used: publish_event(), K9EventBus, BaseMonitor

The Observer pattern notifies interested parties of events without coupling the emitter to the observers. publish_event() decouples agents from the monitoring, messaging, and telemetry systems that consume their events.

25.7 Adapter

Where used: k9_adapters/crewai/

The Adapter pattern converts one interface to another. K9CrewAIAdapter bridges the CrewAI agent interface to K9-AIF's BaseAgent contract, enabling CrewAI agents to participate in K9-AIF squads without modification.

25.8 Builder

Where used: SquadLoader, OrchestratorLoader

The Builder pattern constructs complex objects step by step. SquadLoader builds a BaseSquad by loading YAML, resolving agent classes from the registry, wiring them together, and setting the flow configuration.

25.9 Orchestrator

Where used: BaseOrchestrator

The Orchestrator pattern coordinates multiple services to complete a workflow. BaseOrchestrator is a direct realization of this pattern — it knows the workflow structure and delegates execution to squads and agents.

25.10 Layered Architecture

The overall K9-AIF architecture follows a strict layered approach:

Solution Layer (SBBs)	Domain agents, orchestrators, routers
00B Layer	K9ValidationLoopAgent, K9ModelRouter
ABB Layer (Contracts)	BaseAgent, BaseOrchestrator, BaseRouter
Infrastructure Layer	LLMFactory, persistence, monitoring

Each layer depends only on the layer below it. No upward dependencies.

26. K9X Ecosystem

The K9-AIF framework is part of a four-product ecosystem. Each product is independently installable and deployable.

Product	Purpose	Install	Dependency
K9-AIF Framework	Architecture contracts, agent runtime	<code>pip install k9-aif</code>	Python 3.10+
K9X Studio	Visual architecture designer	<code>pip install k9x</code>	Framework
K9X Enterprise Continuum	SBB/ABB catalog & governance	<code>pip install k9x-continuum</code>	PostgreSQL
K9X HIL	Human-in-the-loop case management	<code>pip install k9x-hil</code>	PostgreSQL, Kafka

Quick Start

Framework only – build and run agents locally

```
pip install k9-aif
```

Framework + visual designer

```
pip install k9-aif k9x
k9x studio
```

Full ecosystem (self-hosted)

```
pip install k9-aif k9x k9x-continuum k9x-hil
```

Product Relationships

- **Studio** generates K9-AIF scaffolds and queries the Continuum for existing SBBs

- **Continuum** stores published SBBs and ABBs; validates compliance via `k9aif inspect`
- **HIL** receives tasks from orchestrators via Kafka; responds to `reply_to` topics
- **Framework** is self-sufficient — runs without any other product

Links

- **Website:** <https://k9x.ai>
 - **Studio:** <https://studio.k9x.ai>
 - **Continuum:** <https://continuum.k9x.ai>
 - **GitHub:** <https://github.com/k9aif/k9-aif-framework>
-

27. Acknowledgements

K9-AIF reflects the accumulated influence of several bodies of knowledge and practice.

Enterprise Architecture

The ABB/SBB distinction is inspired by TOGAF’s Architecture Building Block framework, which distinguishes architectural intent from solution realization. This separation is the foundational insight of K9-AIF.

Software Design Patterns

The Gang of Four design patterns — Template Method, Strategy, Factory, Chain of Responsibility, Observer, Adapter, Builder — appear throughout the framework. These patterns have earned their place because they solve recurring structural problems with proven solutions.

Enterprise Integration Architecture

The Router → Orchestrator → Squad → Agent hierarchy reflects enterprise integration patterns: the routing slip, the process manager, and the message channel. K9-AIF applies these patterns to AI agent workflows.

Agentic AI Systems

The validation loop and actor-critic refinement patterns are informed by the broader agentic AI literature, including ReAct, self-correction, and iterative reasoning research. K9-AIF provides enterprise-grade structure for these patterns.

Open-Source AI Tooling

Ollama, Redpanda, Neo4j, ChromaDB, and the broader Python AI ecosystem provide the infrastructure that K9-AIF orchestrates. The framework is intentionally backend-agnostic at the ABB level.

Claude Code and AI-Assisted Development

Several K9-AIF components — including `BaseValidationLoopAgent`, `BaseCriticActorAgent`, and their OOB implementations — were developed with Claude Code as an accelerator. The experience directly informed Chapter 23 and the guidance on maintaining architecture authority while using AI coding assistants.

Human-Guided Architecture

Every architectural decision in K9-AIF reflects human architectural judgment. AI tools

accelerated implementation; they did not make structural choices. The framework is a record of those choices, not an artifact of automated generation.

28. References

K9-AIF Project

- **Website:** <https://k9x.ai>
- **Main Repository:** <https://github.com/k9aif/k9-aif-framework>
- **Blog:** <https://blog.k9x.ai>
- **Architecture Graph:** <https://graph.k9x.ai>
- **Patterns Repository:** <https://github.com/k9aif/k9aif-patterns>

TOGAF and Architecture References

- The Open Group Architecture Framework (TOGAF) — Architecture Building Blocks and Solution Building Blocks
- *Enterprise Architecture as Strategy* — Ross, Weill, Robertson
- *Software Architecture: Foundations, Theory, and Practice* — Taylor, Medvidovic, Dashofy

Design Pattern References

- *Design Patterns: Elements of Reusable Object-Oriented Software* — Gang of Four (Gamma, Helm, Johnson, Vlissides)
- *Patterns of Enterprise Application Architecture* — Martin Fowler
- *Enterprise Integration Patterns* — Hohpe, Woolf

Agentic AI References

- *ReAct: Synergizing Reasoning and Acting in Language Models* — Yao et al., 2022
- *Self-Refine: Iterative Refinement with Self-Feedback* — Madaan et al., 2023
- *Constitutional AI* — Anthropic, 2022
- *Language Agents as Optimizable Graphs* — relevant agentic orchestration research

Framework Dependencies

- Ollama: <https://ollama.com> — local LLM serving
- Redpanda: <https://redpanda.com> — Kafka-compatible streaming
- Neo4j: <https://neo4j.com> — graph database for architecture topology
- ChromaDB: <https://trychroma.com> — vector database for retrieval
- Docling: OCR and document intelligence tooling

Claude Code

- Claude Code CLI: <https://claude.ai/code>

- Anthropic: <https://anthropic.com>

K9-AIF Developer Guide — Version aligned with k9_aif_abb package
Author: Ravi Natarajan | Project: K9-AIF Framework | <https://k9x.ai>